Program overview

19-May-2025 10:09

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Geomatics**

Code	Omschrijving	ECTS
Master Geomatics		
MSc 1 GM	**MSc 1 Geomatics**	
GEO1000	Python Programming for Geomatics	5
GEO1001	Sensing Technologies	5
GEO1002	Geographical Information Systems (GIS) and Cartography	5
GEO1003	Positioning and Location Awareness	5
GEO1006	Geo Database Management Systems	5
GEO1015	Digital Terrain Modelling	5
MSc 2 GM	**MSc 2 Geomatics**	
GEO1004	3D Modelling of the Built Environment	5
GEO1007	Geoweb Technology	5
GEO1009	Geo-information Governance	5
GEO1016	Photogrammetry and 3D Computer Vision	5
MSc 2 GM - Electives 10 EC	**MSc 2 Geomatics - Electives 10 EC**	
MSc 3 GM	**MSc 3 Geomatics**	
GEO2011	Thesis Preparation	10
Choice of		
GEO1101	Synthesis Project	10
IFM4040	Joint Interdisciplinary Project	15
MSc 3 GM - Electives 10 EC	**MSc 3 Geomatics - Electives 5 or 10 EC**	
MSc 4 GM	**MSc 4 Geomatics**	
GEO2020	Thesis	30
GM Electives	**GM Electives, courses listed below are recommended to be chosen as electives**	
ARFW0501	Applied Spatial Analysis for Sustainable Urban Development	5
ARFW0503	Urbidata: Responsible Design for Fair Data Cities	5
GEO5010	Research Assignment	2
GEO5012	Land Administration	5
GEO5014	Geomatics as support for energy applications	5
GEO5015	Modelling wind and dispersion in urban environments	5
GEO5016	Geomatics in practice	10
GEO5017	Machine Learning for the Built Environment	5
Electives, courses listed below are recommended to be chosen as electives		
AESM5110C	Aerosol and Cloud Microphysics	5
AESM5120C	Data Assimilation for the Geosciences	5
AESM5210C	Climate Remote Sensing	5
AESM5220C	Microwave Remote Sensing of the Earths Surface	5
AESM5230C	Coastal Remote Sensing	5
AESM5240C	Applied Space Geodesy	5
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
AR0916	Beyond 3D Computer Visualisation	5
CESE4055	Ad hoc and Sensor Networks	5
CESE4065	Advanced Practical I.o.T. and Seminar	5
CESE4075	Supercomputing for Big Data	5
EE4740	Data Compression: Entropy and Sparsity Perspectives	5
ET4169	Radar I: From Basic Principles to Applications	5
IN4302TU	Building Serious Games	5
RO47003	Robot Software Practicals	5
SEN9235	Game Design Project	5
WI4227-14	Discrete Optimisation	6

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MSc 1 GM

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part >= 5.75). For assignments that are submitted late, points will be removed, as mentioned in the assignment description. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1001	Sensing Technologies	5
Course Coordinator	Dr. A. Rafiee	
Instructor	Ir. E. Verbree	
Responsible for assignments	Dr. A. Rafiee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Linear Algebra & Calculus at the level of the TUDelft courses Linear Algebra 1 and Calculus 1; basics of statistics (e.g., descriptive statistics, probability distribution and confidence intervals).	
Course Contents	This course provides an overview on the principles and applications of spaceborne, airborne and terrestrial sensing technologies for geographic data acquisition as well as techniques for processing and information extraction from the acquired data. The topics of this course include: Physics of Remote Sensing: Electromagnetic(EM) spectrum and EM interaction with atmosphere and earth surface Satellite imagery data processing and machine learning techniques for thematic information extraction Principles and applications of RADAR and SONAR Principles of Airborne and terrestrial Lidar, introduction to point cloud data acquisition techniques and their applications in different domains Principles of Photogrammetry (acquisition, aerotriangulation, adjustment, Stereophotogrammetry, orthophoto and quality assessment) Introduction to positioning systems Introduction on preparing sensor outputs for GIS storage: (1) image matching techniques; (2) digital image processing; (3) machine learning for automated information extraction and quality assessment approaches including (4) least squares adjustment (LSA); (5) surface fitting for outlier detection.	
Study Goals	After the course the student is able to describe different sensing techniques, from different platforms, for acquiring geospatial data explain the applied metrics for quality assessment of the acquired data apply data processing and machine learning techniques on the (self) acquired geographic data analyse the applied data processing/ information extraction algorithms and evaluate their capabilities/drawbacks compare different sensing techniques for their suitability in different application domains from their efficiency and quality points of view select appropriate sensing technologies for tackling a spatial decision making problem	
Education Method	Lectures 30 hours; 26 hours Assignments; 84 hours self-study. The assignments aim to support gaining insight into the diverse techniques, methods and concepts explained during the lectures. Students will also participate in a 2-days excursion to Intergeo, the biggest geomatics event worldwide. In 2024 the event will be held on 24 26 September in Stuttgart, Germany (see further: https://www.intergeo.de/en/). Students are recommended (but not obliged) to visit Intergeo on-site.	
Books	Handouts and open books will be provided to the students	
Assessment	Written exam and Assignments. The result of the written exam determines 70% of the final score and the Assignments 30%. An overall grade of 5.75 or above is required to pass the course. Assignments submitted after the deadline: the grade will be deducted with 10% per day late, and not accepted after 3 days late. Only students that failed an assignment (grade lower than 5.75) are eligible for a repair task. A successful repair task will result in maximum grade of 6,0 for that assignment.	
Period of Education	Quarter	
Leerstoel	GIS Technologie	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002 Geographical Information Systems (GIS) and Cartography 5

Course Coordinator
Dr. G. Agugiaro

Instructor
Dr. D. Cannatella

Instructor
Dr. G. Agugiaro

Instructor
A. Petrovi

Responsible for assignments
Dr. G. Agugiaro

Contact Hours / Week x/x/x/x
6 hours per week

Education Period
1

Start Education
1

Exam Period
1
2

Course Language
English

Expected prior knowledge
This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (e.g., architecture, urban planning, civil engineering).

However, prior knowledge of scripting/programming in at least one language (e.g., Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).

Course Contents
The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.

The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.

The course has 3 parts:

1. Introduction to GIS
- fundamentals of Geodesy, Coordinate Reference Systems, and map projections,
- spatial data modelling (vector and raster spatial models),
- geo-data manipulation (editing, digitizing, importing, converting, etc.),
- overview of spatial analysis operations,
- production of interpretable output (e.g. maps),
- fundamentals of data quality.

2. Algorithms and data structures for GIS
- data structures for vector and raster data (including topological data structures),
- basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.),
- basic algorithms for raster (encoding, quad trees, map algebra),
- networks and related algorithms such as shortest-path.

3. Applications of GIS to real-world problems
Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.

Study Goals
After the course Introduction to GIS and digital cartography the student will be able to:
1) Explain what a GIS is and what real-world problems it can help solve;
2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;
3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;
4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved;
5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;
6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.

Education Method
Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours

Literature and Study Materials
- Book: Principles of Geographical Information Systems
- Slides of the lectures (available on Brightspace)
- Additional selected book chapters or scientific articles (available on Brightspace)

Assessment
Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.

The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.

A student passes the course only if the following two conditions are satisfied:
1. The average grade of both graded assignments is >= 5.75.
AND
2. The grade of the final exam is >= 5.75.

If a student does not pass for the graded assignments, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it.

The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.

Period of Education
Quarter 1

Course evaluation
For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO1003	Positioning and Location Awareness	5
Course Coordinator	Ir. E. Verbree	
Instructor	Dr. A. Rafiee	
Responsible for assignments	Ir. E. Verbree	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course addresses Global Navigation Satellite Systems (GNSS) and (indoor) positioning technologies for sensing people, devices, and assets in the built environment with the focus on location-aware applications. The course covers the requirements and context for these location-aware applications: global, local, and linear reference systems, coordinate systems and map projections, positioning methods and techniques, and the social and technical push and legislative pull factors that empower the development of location-based services. The course consists of three parts: 1. Analyses of location aware applications: concepts of location awareness, location sensitivity, and context awareness; use cases of location-based applications; technical and legislation factors, privacy issues, and standards. 2. Global, local (e.g. Dutch), and linear reference systems; coordinate systems and transformations; basic types of map projections. 3. Positioning methods, techniques, and performance: Global Navigation Satellite Systems; terrestrial radio-based positioning by using Bluetooth, UWB, RFID, Wi-Fi and other short range technologies; Inertial Navigation Systems (INS); presence-based locating; technologies with dead reckoning and map-matching methods; fingerprinting (templates) for indoor positioning.	
Study Goals	After the course Positioning and Location Awareness the student is able to: 1. Understand location awareness, location sensitivity, context awareness; 2. Understand the different types of reference systems: global, local (Dutch), linear; 3. Understand the ethical and legislative factors of methods for location awareness; 4. Apply different coordinate systems, positioning, and indoor localisation; 5. Evaluate different technologies to support location awareness on their technical performance (availability, accuracy, integrity, continuity), and their ethical factors and legislative factors (privacy issues).	
Education Method	Lectures: 28 hours Assignments: 72 hours Self-study: 40 hours	
Assessment	- Group Assignments (20%) - Individual Assignments (20%) - Written Exam (60%) Students need to get at least a sufficient grade (5.75) for both the exam and the assignments in order to obtain the overall grade for the course. The grades for both group and individual assignments are recorded on Brightspace. Should an assignment receive a grade lower than 5.75, students have the opportunity to improve it through revision or addition. A successful improvement will result in a reassessment with a 6.0. If a student scores 5.75 or lower on the written exam, they may retake it during the next available opportunity.	
Period of Education	Quarter	
Used Materials	Reader	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1006	Geo Database Management Systems	5

Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom
Course Coordinator	Drs. C.W. Quak
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom
Contact Hours / Week x/x/x/x	2*2 hours lectures and 3 hour (assignment support + feedback)
Education Period	2
Start Education	2
Exam Period	2 3
Course Language	English
Expected prior knowledge	Wiskunde B VWO (Mathematics B of Dutch VWO education) Time/space complexity of algorithms Binary number representation Operating system command line (e.g. of Windows, Linux) Geomatics courses: GEO1002 (GIS and Cartography) and GEO1000 (Python Programming)
Course Contents	This course is about managing geo-information in a database management system (DBMS). The geo-DBMS is the central component of the geo-information chain. All information observed and interpreted should be managed in a manner that supports further use, so the Geo-DBMS is the end-point of data collected. Analysis, visualizing and (web-) dissemination of the geo-information start from the geo-DBMS. It is crucial that the content of the geo-DBMS is clear (the information model and its associated semantics) to the users, but also that it is high performant, as the geo-data volumes are often very high and especially in a web-based environment there may be many uses. The course consists of three parts; the first two concern introductions. The main part is devoted to spatial modelling and geo-DBMS. More specifically the three parts cover: 1. Introduction object-oriented information modelling: UML and more specifically class diagram to create conceptual models. Mastering (and applying in design) the key notions such as: Class, Abstract Class, Identifiers, Instances, Attribute, Operation, Association, Generalization, Aggregation, Composition, Multiplicity, Note, Constraint, Package, Stereotype, etc. 2. Introduction relational database management systems: the relational model, SQL DDL (schema definition) and DML (queries). Relational algebra. Some implementation issues (clustering and indexing techniques, such as the b-tree, primary and foreign key constructions supporting referential integrity). In addition a first introduction to NoSQL (Not Only SQL) databases. 3. Spatial databases: theoretical aspects of spatial modelling (object and field models), DBMS-GIS architectures (dual, layered, integrated), spatial-temporal data model, Abstract Data Types (ADTs) for geometric primitives, topology model (ISO 19107), temporal aspects, spatial indexing (quad tree, r-tree, field-tree), spatial data clustering (Hilbert, Morton, Cantor, row, row prime, etc.), geo-DBMSs (data modelling, structures, queries, visualisation), (non-)commercial systems (Oracle Spatial, PostGIS, etc.). One or more advanced topics: 3D, Raster, Point cloud data, Simplicial complexes, Geo-OCL, Big data, NoSQL (spatial), Vario-scale, 5D modelling (deeply integrated 3D space, time and scale).
Study Goals	After the course the student is able to: - design an conceptual information model by converting a descriptive text of a real world situation into a Unified Modelling Language (UML) class diagram. - create a relational database management system (DBMS) schema to store the information for a real world situation (as captured in a UML class diagram), including definition of (primary/foreign) keys, clustering and indexing for performance. - apply the Structured Query Language (SQL) to query and update a relational DBMS by using a range of techniques: join two or more tables, aggregate data, specify meaningful selection predicates, ordering the output, use subqueries and program additional functionality with a procedural language (PL/pgSQL in PostgreSQL). - understand the different characteristics of spatial data and be able to also design a conceptual model for a spatial DBMS, create the spatial conceptual information model in a RDBMS with spatial extensions, optimize the implementation of spatial databases (spatial clustering indexing, spatial constraints,...) and retrieve and update spatial data using spatial operations (both geometric and topological) in the selection in combination with non-spatial predicates. - apply a range of advanced topics: spatial-temporal modelling, 3D modelling, routing inside the DBMS (using pgRouting) vario - or multiscale modelling, Spatial OCL (Object Constraint Language) formalization, simplicial homology, nD point clouds, efficient raster data management, 5D modelling, and selected NoSSQL database (Neo4j, MongoDB).
Education Method	Lectures per week: 4 hours theory and 3 hour (assignment support + feedback). Lectures: 28 hours Assignments (supervised): 21 hours Self-study: 91 hours (includes reading and practical assignments, unsupervised parts) The assignments require the use of software: Enterprise Architect, PostgreSQL/PostGIS and Quantum GIS.
Literature and Study Materials	- Textbook (Elmasri & Navathe, Fundamentals of Database systems, 5th. Int. Edition, Addison-Wesley, ISBN: 0-321-41506-X). - PostgreSQL user manual - PostGIS manual - Randy Miller, Practical UML: A Hands-On Introduction for Developers. - Scott W. Ambler, UML Class Diagrams, excerpt from: The Object Primer, 3rd Edition: Agile Model Driven Development with UML 2 - Tom Vijlbrief and Peter van Oosterom (1992). GEO++: An extensible GIS. In proceedings 5th International Symposium on Spatial Data Handling, Charleston, South Carolina, pages 40-50, August 1992. - Section 6.1 (Geometry object model, esp. 6.1.15 Relational operators) and section 7.2.6 (Examples) of OGC SFS (2011), that is, the OpenGIS® Implementation Standard for Geographic information - Simple feature access - Part 1: Common architecture. - Eliseo Clementini, Paolino Di Felice and Peter van Oosterom. A Small Set of Formal Topological Relationships Suitable for End-User Interaction. The 3rd International Symposium on Large Spatial Databases, Singapore, (LNCS nr. 692), pages 277-295, June 1993. - Peter van Oosterom, Wilko Quak and Theo Tijssen (2004). About Invalid, Valid and Clean Polygons. In: Peter F. Fisher(Ed.); Developments in Spatial Data Handling, 11th International Symposium on Spatial Data Handling, 2004. - Peter van Oosterom (2006). Constraints in Spatial Data Models, in a Dynamic Context. In: J. Drummond, R. Billen, E. Joao and D. Forrest (Eds.); Dynamic and Mobile GIS: Investigating Changes in Space and Time, pp. 104-137 - Peter van Oosterom. Spatial Access Methods. Chapter T2.3 in Geographical Information Systems Principles, Technical Issues, Management Issues, and Applications (edited by Longley, Goodchild, Maguire en Rhind), Wiley pages 385-400 (vol.1), 1999. - Peter van Oosterom and Tom Vijlbrief. 'The Spatial Location Code', presented at SDH'96, Delft 12-16 August 1996. - P.J.M. van Oosterom, B. Maessen and C.W. Quak (2002). Generic query tool for spatio-temporal data. In: International Journal of Geographical Information Science, Volume 16, 8, 2002, pp. 713-748 - Peter van Oosterom, Hendrik Ploeger, Jntien Stoter, Rod Thompson and Christiaan Lemmen (2006). Aspects of a 4D Cadastre: A First Exploration. XXIII International FIG congress, Munich, October 2006, 23 p. - Friso Penninga and Peter van Oosterom (2008). A simplicial complex-based DBMS approach to 3D topographic data modeling. In: International Journal of Geographical Information Science, Volume 22, 7, 2008, pp. 751-779. - Peter van Oosterom (2005). Variable-scale Topological Data Structures Suitable for Progressive Data Transfer: The GAP-face Tree and GAP-edge Forest. In: Cartography and Geographic Information Science, Volume 32, 4, 2005, pp. 331-346 - Peter van Oosterom and Martijn Meijers (2014). Vario-scale data structures supporting smooth zoom and progressive transfer of 2D and 3D data. In: International Journal of Geographical Information Science, Volume 28(3), 455478. - Lina Huang, Martijn Meijers, Radan uba, Peter van Oosterom (2016). Engineering web maps with gradual content zoom based on streaming vector data, In: ISPRS Journal of Photogrammetry and Remote Sensing, Elsevier BV, 114, pp. 274-293, 2016. - Peter van Oosterom, Oscar Marinez-Rubi, Milena Ivanova, Mike Horhammer, Daniel Geringer, Siva Ravada, Theo Tijssen, Martin Kodde, and Romulo Goncalves (2015). Massive point cloud data management: design, implementation and execution of

	a point cloud benchmark. In Computers&Graphics. - Peter van Oosterom, Oscar Martinez-Rubi, Theo Tijssen, Romulo Gonçalves (2016). Realistic Benchmarks for Point Cloud Data Management Systems, Chapter in: Advances in 3D Geoinformation, Springer Nature, pp. 1-30.
Assessment	Written exam (3 hours) and lab assignments. The exam result will determine the final score, but lab assignments need to be sufficient (>=5,75). The main purpose of lab assignment is to support the understanding of the theory via hands-on experience. The exam is the only summative assessment and determines your grade for 100%. However, students need to pass at least five of the formative assignments (lab assignment) in order to pass the course. In week 8 of the course, there is one extra formative assignment for students that did not pass all five assignments.
Period of Education	Second Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO1015	Digital Terrain Modelling	5
Course Coordinator	H.F. Ledoux	
Instructor	Dr.ir. G.A.K. Arroyo Ohori	
Responsible for assignments	H.F. Ledoux	
Contact Hours / Week x/x/x/x	6h/week	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Digital terrain models (DTMs) are computer representations of the elevation of a given area, and they play an important role in understanding and analysing our built environment. They are the necessary input for several applications (eg flood modelling, visibility, effects of climate change on the north poles, etc.), and they are also relevant for studying the seabed. The course provides an overview of the fundamentals of digital terrain modelling: - different representations of terrains: TINs, rasters, point clouds, contour lines - reconstruction of terrains from different sources (lidar, InSAR, photogrammetry, multibeam echosounders) - spatial interpolation methods - conversion between different representations - processing of terrains and point clouds: outlier detection, filtering, segmentation, and identification and classification of objects - techniques to handle and process massive datasets - applications, eg runoff modelling, watersheds computations, visibility The course has both a theoretical part and a practical part where students reconstruct, manipulate, process, and extract information from terrains. All the labs are programming tasks (to be done with Python and/or C++), based solely on open-source libraries and software.	
Study Goals	At the end of the course, students will be able to: - describe the pros and cons of different representations of terrains - explain how elevation datasets can be automatically converted to terrains - reconstruct and manipulate terrains using with open-source libraries - analyse how terrains can be useful in different applications related to built environment - given a specific problem where elevation plays a role (eg visibility or flood modelling), analyse and identify which data and algorithms are needed to solve the problem, and assess the consequences of these choices	
Education Method	We use a "flipped classroom": each week there are 2 main topics, and students first watch the videos and read the material at home. Then during the contact hours (3 x 2h), the most difficult parts are discussed and students get help/support for the practicals.	
Books	The open book "Computational modelling of terrains", written by the instructors of the course, is used. Freely available at https://tudelft3d.github.io/terrainbook/	
Prerequisites	The course is designed for students from the MSc Geomatics, and the following courses are required prerequisites: (1) GEO1000 (or knowledge of scripting/programming in at least one language (eg Matlab, C++, or Python) (2) GEO1001 (3) GEO1002 Students from other faculties with the appropriate background are welcome.	
Assessment	- two written exams (1 mid-term (at week 2.5) + 1 final exam) - assignments (programming in Python and/or C++) - there are 2 retakes (during Q3): one for both written exams + a large retake assignment covering all the assignments The weights for the exams and the assignments are around 50%/50%. Assignments submitted after the deadline: the grade will be deducted with 10% per day late, and not accepted after 3 days late. A mark of 5.00 in the exams part and a total course mark of 5.75, are both necessary to pass the course.	
Remarks	All material can be found at https://3d.bk.tudelft.nl/courses/geo1015/ Brightspace will not be used.	
Period of Education	Q2	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Geomatics**

MSc 2 GM

GEO1004	3D Modelling of the Built Environment	5
Course Coordinator	Dr.ir. G.A.K. Arroyo Ohori	
Instructor	H.F. Ledoux	
Responsible for assignments	Dr.ir. G.A.K. Arroyo Ohori	
Contact Hours / Week x/x/x/x	Read 6 hours per week starting from week 3.1 and ending in week 3.7	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	The course is mainly designed for students from the MSc Geomatics, but everyone is welcome to follow the course. However, basic knowledge of GIS is recommended, and some knowledge of scripting/programming (in any language) is required to do the assignments. Within Geomatics, these are covered by the courses GEO1002 and GEO1000.	
Course Contents	This course provides a detailed description of the main ways in which the built environment is modelled in three dimensions, covering material from low-level data structures for generic 3D data to high-level semantic data models for cities. The main topics described in the course are: - how geometry, topology and attributes can be stored in 3D models; - boundary representation (b-rep) and meshes; - (constrained) tetrahedralisation and 3D Voronoi diagrams; - voxel models and voxelisation; - constructive solid models (CSG) and Nef polyhedra; - Bézier surfaces; - the medial axis transform (MAT); - generalised (g-) and combinatorial maps (c-maps); - the ISO 19107 geometry representations; - semantic 3D city models and CityJSON; - building information modelling and IFC; - building reconstruction for 3D city models; - applications of 3D models (eg calculation of volumes and solar potential); The course has both a theoretical part based on short lectures, watching videos and reading the provided material, as well as a practical part where students create, manipulate and view 3D models, and then use them for different applications. All the labs are programming tasks (to be done with the C++ programming language), based solely on open-source libraries and software.	
Study Goals	At the end of the course, students should be able to: 1. compare different 3D modelling approaches, outline their relative merits and drawbacks, and choose an appropriate approach for a given use case; 2. implement several different data structures for the storage and processing of 3D models; 3. create and manipulate data sets that use the main semantic open data models used in GIS (CityGML-CityJSON) and BIM (IFC); 4. execute analyses based on a 3D city model and check their result.	
Education Method	Blended-learning: Each week there are 2 main topics. During the contact hours, there are short lectures and students get help/support for the assignments. Outside the contact hours, the students work on their group assignments, watch videos and read the provided study material on their own.	
Literature and Study Materials	The course follows the 3D book, which is specifically developed for this course: https://github.com/tudelft3d/3dbook	
Assessment	- Two written examinations: midterm (5%) + final exam (45%) - Three C++ programming assignments (10%+20%+20%), which are practical exercises combining multiple course topics A mark of 5.00 in the final exam and a total course mark of 5.75 are both necessary to pass the course. There is a retake for every assignment and a combined retake for the exams.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: - Course enrolment via Brightspace only is not considered a valid registration - Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) - The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	All material can be found at https://3d.bk.tudelft.nl/courses/geo1004/ Brightspace will not be used.	
Period of Education	Quarter 4	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1007	Geoweb Technology	5
Course Coordinator	Dr.ir. B.M. Meijers	
Instructor	Dr. A. Rafiee	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	Wiskunde B VWO (Mathematics B of Dutch VWO education) Familiarity with operating system command line (e.g. of Windows, Mac, Linux) Geomatics courses: GEO1000 (Python Programming), GEO1001 (Sensing Technologies), GEO1002 (GIS and Cartography), GEO1003 (Location Awareness), GEO1006 (Geo-DBMS) and GEO1009 (Geo-Info Law)	
Course Contents	Today professionals and laymen alike have access to geo-data through the internet. Using a smart phone or tablet connected to broadband internet one can virtually fly-through a city and view buildings, underground constructions and proposed designs from any viewing angle, even augmented reality. The internet also strengthens the communication between government and citizens, who can comment on the designs, fill in questionnaires and vote, and in this way (local) government may increase public participation in city planning and other public affairs. Access to geo-web services should be fast & reliable and based on open standards such as OGC (Open Geospatial Consortium), ISO (International Organization for Standardization) and INSPIRE (European Spatial Data Infrastructure). The map interfaces should be user friendly while sufficient functionality should be available for carrying out analyses on the diverse datasets. A key tool when building geo-web services is Geography Markup Language (GML) an exchange language for geo-data based on XML (eXtensible Markup Language), one of the standard languages of the internet. The course consists of four parts: 1. Principles, application, evaluation and integration of generic web services and the importance of using standards when building them. The standards involved start with general ICT standards like HTTP, XML and JSON. Increasingly the semantics of data in the framework of web services, and linked data, plays an important role. 2. Based on these general standards, geo-standards and protocols can be build (ISO TC-211/OGC web services and protocols). Combining the geo-standards and protocols results in a set of geo web services that more and more replace the traditional GIS tools. Topics of interest: geo-web system server-client architecture, WMS/WFS/... server, desktop/browser/mobile clients (incl. web application development), web-based transaction processing, portrayal, query (filter encoding), tiling, metadata service, web processing service, point cloud & vario-scale data services. 3. Principles and applications of (real-time) Sensor Web. The components of Sensor Web will be explained and (some of them) used: Observations & Measurements, Transducer Markup Language, Sensor Observation Service, Sensor Planning Service, Sensor Alert Service and the Web Notification Service. 4. Applying software tools for the visualization of 3D geo-data. Normally the first, and in many cases the most important, thing to do with (3D) geo-data is to visualize it. Nowadays data is delivered over the internet by means of web services. The standards, architecture and services for 3D geo-visualization, and the practical aspects of available (web) tools and how to use them will be explored.	
Study Goals	After the course the student is able to: - understand generic internet and web standards and apply these in the implementation of web information systems - describe the existing formal geo-standards, derived implementation specifications and related geo web services and their application domains - understand the Sensor Web Enablement Common Data Framework (common data models and schema) and the SensorML (models and schema for sensor systems and processes surrounding measurements) and use them in web services - describe the available (visualization) standards for the visualization of 3D geo-data, like X3D, 3D-pdf, WebGL, CityGML, KML, GeoJSON, glTF, b3dm, i3s and other industry standards / software tools - integrate several geo web services for supporting spatial decision making - evaluate existing geo web services in terms of specified applications and give motivated suggestions for improvement	
Education Method	Lectures (including guest lectures from practice): 28 hours Self-study: 48 hours Lab assignments (64 hours): on the various internet, web, geoweb standards, protocols, services and tools	
Literature and Study Materials	Selected chapters, standards documents & scientific and professional articles made available via Brightspace.	
Assessment	Written exam (3 hours), and lab assignments. The exam result will determine 60% of the final score, the lab assignments make up the other 40%. To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part >= 5.75, where missing grades are 0). For assignments that are submitted late, points will be removed. A repair option is offered for one of the lab assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam For the exam, a re-take option is available.	
Period of Education	Fourth Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1009	Geo-information Governance	5
Course Coordinator	Dr.ir. B. van Loenen	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	Dr.ir. B. van Loenen	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In this course students will learn about the organisational and legal aspects relevant for developing a strategy for a spatial data infrastructure. Many organisations, and increasingly also citizens, collect, process and disseminate geo-information (e.g., Google, TomTom, mapping agencies and cadastres). Some information serves a specific purpose, while other information acts as multipurpose or base information, which is the fundament of an information infrastructure. An adequate information infrastructure allows for information to be collected efficiently (collect it once, use it many times) and provides reliable information for effective use in decision-making processes at all levels in government, private sector, academia and among citizens. Legislation is a critical component that needs to be considered to determine whether information can be used for a certain purpose and shared with others. The course consists of three parts: 1. Theoretical basis for the introduction, organisation and management of (spatial) data infrastructures. Aspects that will be addressed are stakeholders, co-ordination mechanisms and components of a data infrastructure, the affiliation with policy lines like the European Green Deal and Data Spaces. 2. Overview of legislation pivotal to developing geographic information infrastructures. Legal aspects applying to geo-information, such as intellectual property rights, privacy and open data legislation, constitute a major part of this course. 3. Practical application of the theory. Students are tasked to write a paper on a selected SDI topic (group assignment).	
Study Goals	After this course the student is able to: - recognize and anticipate upon relevant legal and organisational issues related to the acquisition, processing, dissemination and use of (open) geo-information - apply the concepts, processes and main components of geo-information infrastructures to support geo-information sharing between organisations - critically assess geo-information management strategies for organisations - assess the performance of a geo-information infrastructure from a user perspective - (co-)author a scientific paper on a selected SDI topic	
Education Method	Interactive lectures/ workshops: 28 hours; Group assignment: 42 hours; Self-study: 70 hours.	
Assessment	Exam; Group assignment (article, peer review and presentation). The exam will determine 50% of the final score and the group assignment (writing an article) also 50%. Students need to get at least a 5.75 grade for both the exam and the group assignment in order to pass the course. The mark for the group assignment can be influenced for 0.5 grade point (maximum) by the individual reflection on the article. Group assignments submitted after the deadline: the grade will be deducted with 10% per day late, and not accepted after 3 days late. Students need to (1) present preliminary results of their research, (2) review the article of another student (group), and (3) take part in the final debate in order to obtain a final grade for the group assignment. There is one retake for the exam. Students that fail the group assignment, can repair the group assignment. A successful repair will obtain a maximum grade of 6.0. Deadline for the repair assignment is 1 June of the respective study year.	
Period of Education	Quarter 3	
Leerstoel	Urban Data Science	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1016	Photogrammetry and 3D Computer Vision	5
Course Coordinator	Dr. L. Nan	
Contact Hours / Week x/x/x/x	6 hours/week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Photogrammetry and 3D computer vision aim at recovering the structure of real-world objects/scenes from images. This course is about the theories, methodologies, and techniques of 3D computer vision for the built environment. In the term of this course, students will learn the basic knowledge and algorithms in 3D computer vision through a series of lectures, reading materials, lab exercises, and group assignments. The topics cover the whole pipeline of reconstructing 3D models from images: - Cameras models: how a point from the real world gets projected onto the image plane and how to recover the camera parameters from a set of observations; - Epipolar geometry: the geometric relations between 3D points and their images points; the constraints between the image points; - Image matching: define and match image features (SIFT) to establish correspondences between images; - Structure from motion: recover/refine geometry and camera parameters from a set of images; - Multi-view stereo and learning-based approaches for recovering dense geometry (e.g., point clouds) from images; - Surface reconstruction: obtain 3D surface models of real-world objects from point clouds.	
Study Goals	After finishing this course, the students will be able to: - apply linear algebra knowledge to implement basic 3D computer vision algorithms; - explain the main concepts in 3D computer vision (i.e., camera models, epipolar constraints, fundamental matrix, image matching, triangulation, structure from motion, bundle adjustment, multi-view stereo, and surface reconstruction); - explain the principles of the state-of-the-art 3D computer vision pipelines for 3D dense reconstruction from images; - evaluate methods for reconstructing smooth surfaces and piecewise planar objects, and choose applicable methods to solve specific reconstruction problems; - propose and implement solutions for reconstructing real-world buildings from images.	
Education Method	Lectures, reading materials, group assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is related to all fields of Geomatics for the Built Environment, from data acquisition, data management, data visualization, data analysis, 3D modeling, to applications.	
Literature and Study Materials	In addition to the lectures and handouts (distributed after each lecture), you will find relevant topics discussed in great detail in the following materials. [1] Y. Ma, et al. An Invitation to 3-D Vision: From Images to Models. 2001 [2] R. Hartley and A. Zisserman. Multiple View Geometry in Computer Vision. Second Edition. Cambridge University Press, 2004. [3] D. Forsyth and J. Ponce. Computer Vision: A Modern Approach. Second Edition. Pearson, 2012. [4] R. Hartley and P. Strum. Triangulation. Computer vision and image understanding. 1997. [5] Berthold K.P. Horn. Recovering baseline and orientation from essential matrix. 1990. [6] Y. Furukawa and J. Ponce. Accurate, Dense, and Robust Multiview Stereopsis. PAMI, 2010 [7] H. Hoppe, et al. Surface reconstruction from unorganized points. SIGGRAPH, 1992 [8] M. Kazhdan, M. Bolitho, H. Hoppe. Poisson surface reconstruction. SPG, 2006, 61-70 [9] L. Nan and P. Wonka. PolyFit: Polygonal Surface Reconstruction from Point Clouds. ICCV 2017.	
Practical Guide	In the assignments and group work sessions, students will experiment with the 3D computer vision techniques introduced in the lectures. - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; - Discuss the effect of parameters (if possible); - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Basic linear algebra and calculus; - Basic knowledge and skills in C++ programming. Students who have no experience in the above topics can still participate but should be willing to make efforts and invest some more time.	
Assessment	The assessment of this course consists of three group assignments and the final written exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded lower than 5.75 is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.	
Period of Education	Quarter 3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Geomatics**

MSc 2 GM - Electives 10 EC

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Geomatics**

MSc 3 GM

GEO2011	Thesis Preparation	10
Course Coordinator	Dr. C. Garcia Sanchez	
Responsible for assignments	Dr. C. Garcia Sanchez	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The graduation work is the last step in the Geomatics education. It is completed in the form of an individual research project. GEO2020 follows GEO2011, and involves carrying out the research according to the plan of GEO2011. The topic of the scientific project can be related to any of the subjects offered by the Geomatics programme. The project must be supervised by one or more of the lecturers who participated in the Geomatics education. The project can be performed in cooperation with a company/institution, which provides a use case or a problem statement (but the work is always supervised by the scientific staff of the university). The graduation work deliverables are: (1) a scientific report (a thesis); and (2) an oral presentation.	
Study Goals	During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research project in the field of Geomatics. On completing the Graduation Project, the student is able to: 1. Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research; 2. Formulate a theoretical, numerical and/or experimental framework and delineate a research problem such that it can be solved; 3. Interpret obtained results in a critical manner; 4. Explain the work performed in a structured report that incorporates verification of methods and tools and is written in correct English; 5. Communicate the work performed in a structured way through an oral presentation to a wider audience; 6. Demonstrate capacity to manage the project, both technically and time-wise, considering resources and methodology;	
Education Method	Individual work.	
Practical Guide	All details are available at: https://3d.bk.tudelft.nl/courses/geo2020/ Brightspace is not used.	
Assessment	scientific report + presentation. These are evaluated by the graduation committee, guided by the geomatics graduation manual (i.e. the P2 assessment).	
Special Information	Geomatics students have the possibility to carry out their graduation research project in collaboration with a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found at https://3d.bk.tudelft.nl/courses/geo2020/company/. The agreements can be signed at the secretariat of Education and Student Affairs.	
Period of Education	All.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Geomatics**

Choice of

GEO1101	Synthesis Project	10
Course Coordinator	Ir. E. Verbree	
Course Coordinator	Dr.ir. B. van Loenen	
Course Coordinator	H.F. Ledoux	
Instructor	Ir. E. Verbree	
Instructor	H.F. Ledoux	
Responsible for assignments	Dr.ir. B. van Loenen	
Contact Hours / Week x/x/x/x	X/0/0/X	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	GEO2011/GEO2020	
Expected prior knowledge	Geomatics students should have gained 30 EC of core geomatics courses (Core courses are: GEO1000, GEO1001, GEO1002, GEO1003, GEO1004, GEO1006, GEO1007, GEO1009, GEO1015, and GEO1016). External students are admitted if they have similar knowledge and skills based on a certified course list.	
Course Contents	The project aims at investigating a real-world geo-information problem. Geomatics teachers in cooperation with a company, governmental agency, or a TU Delft research group that is not involved in the Geomatics program the clients define the problem. The problem will be addressed by a group of students. The project enables you to gain further knowledge and skills in acquisition, processing, analysis and interpretation, and visualization of geo-information as well as insights in the running of a project in which diverse stakeholders are involved.	
Study Goals	After this course students are able to: - understand, through a project at hand, how each of the individual parts of the geo-information process (from measurement (geo -data gathering), geo-data processing and analysis, to presentation, delivery and application) relate to each other and to the chain as a whole. - apply and expand the knowledge and competences acquired during the core Geomatics courses by developing a solution to a real world geo-information problem. - apply the basic components of a project management strategy to a project addressing a real world geo-information problem. - cooperate and communicate with students, tutors and geo-information professionals. In the cooperation, the student shows commitment, perseverance, responsibility, sense of perspective and leadership. - critically reflect on the quality of the project result(s), on the feasibility to implement the project result(s) in practice, and on their own performance in the project. - present the work performed in a structured way through an oral presentation to their peers and the wider geo-information community.	
Education Method	Group Project	
Course Relations	Geomatics Core Courses: GEO1000, GEO1001, GEO1002, GEO1003, GEO1004, GEO1006, GEO1007, GEO1009, GEO1015, and GEO1016.	
Reader	Project Guide	
Assessment	Your final grade is compiled from a group grade for the project final report. The group grade is based on the technical quality of the project, and the performance of the student in the research process. The individual reflection on the final report can influence positively or negatively for 0,5 grade point (maximum) the grade on the final report. The grading is detailed in the rubrics available on Brightspace. In order to pass the course, students also have to attend and take part in the group's midterm presentation, and two final presentations (one technical presentation for Geomatics staff/ students and one outreach presentation on Geomatics Day), and do the team peer review (on both the midterm report and final technical presentation). To pass the course, a group mark and an individual mark of at least a 5.75 is required. Students that fail the course have to do a repair assignment for a maximum grade of a 6.0 for the course.	
Enrolment / Application	BIS: only for Geomatics students (and Approved International 'Study Abroad' Students).	
Special Information	Deliverables are: - Project Identification Document (PID) (group task) - Mid-term report including proposed amendments to the PID if necessary (group task) - Mid-term reflection on the project including the mid-term report, group process and individual performance (individual task) - Oral mid-term presentation, including status and progress of the project - Final written report: A template will be provided to obtain uniformity in the structure of all synthesis reports (group task) - Reflection on the project including the final report, group process and individual performance (individual task) Oral reporting during Geomatics Day: - Presentation by the team including discussion (group task). You present your project to an extended audience of clients, tutors, peers, and interested professionals.	
Period of Education	Quarter	
Concept Schedule	A kick-off meeting will be arranged at the end of Q4 (the quarter preceding the Q1 of the GEO1101 course concerned). This plenary session has to be attended by all students who want to carry out the synthesis project. The geomatics teachers who are your tutors will be present and, if possible, clients will be present/involved. The client or tutors will present the use case. In consultation with the tutors the coordinator will assign a project to you. In the first week of Q1 each project team and their tutors(s) and/ or the client will come together to discuss the use case.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

IFM4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Instructor	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	To enter JIP, a student should be a 2nd-year master student; have completed the first year of the masters degree programme and will enrol in the second year of the masters degree programme at TU Delft. be full time available (40 hours per week) in Q5; meet the application criteria: JIP application form, JIP motivation letter + top 3 company cases, CV Students will be selected by the JIP core staff based on motivation, study progress and recommendation.	
Course Contents	The Joint Interdisciplinary Project aims to prepare students to contribute to impactful technological solutions for industrial/governmental R&D and support one or more of the Global Sustainable development Goals or ToPsectoren in the Netherlands. Interdisciplinary and Intercultural teams of students work a dedicated ten weeks on a complex problem to realise an innovative and viable (primarily)technical solution. Team deliverables include a report, prototype, concept model and a business case to instigate continued developments. A company coach and JiP staff provide team guidance. Along the way, the companies and TU Delft offer students academic and industry expertise. The projects demand sound engineering working knowledge and experience with interdisciplinary and systems theory, knowledge and mindsets of innovation and entrepreneurial behaviour. The project brief is provided by renowned companies like Airbus, Embraer, Huisman,VanOord, Royal Haskoning DHV, Tahmo, Nationale Politie, Erasmus MC, and KLM.	
Study Goals	Meta-cognitive abilities attributable to interdisciplinary learning LO 1: Demonstrate disciplinary grounding, showing synthesis and integration of different disciplinary scientific and professional (technological) knowledge systems and critical evaluation skills of the approaches used to solve complex problems (Ilo;K) Scientific and intellectual development LO 2: Demonstrate the ability to carry out the analysis and evaluation of ethical, scientific and societal consequences of the proposed innovation (ILO: I) Research and design capabilities LO3: Demonstrate the ability to create reasonable and relevant output, in accordance with the academic standards of well conducted research, design, modelling and technical approaches for the problem addressed within JIP. (ILO: I,J) Collaboration and communication in an interdisciplinary team LO 4: Demonstrate behavioural competences and skills relevant for interdisciplinary teamwork and effective communication with different stakeholders (ILO: K) Self-adjustment and reflection capabilities LO 5: Demonstrate to carry out regular reflections on professional and personal development and being able to improve upon those reflections (ILO: K)	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	Students will meet (online) during the module and discuss their project with professionals, the company, and academia. The JIP core team will also guide the team process. The module includes three review sessions: Problem statement review the teams are required to present the scope of their problem in the form of a two-pager report. The main activity during this review is the team presentation of the academic staff, company coaches, JIP core staff and teams within the same cluster. The formative and 360 grades feedback means giving a head start in the execution of the project. The input is based on a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. Midterm review The teams present multiple solution concepts in a four-page report and a presentation to the academic staff, company coaches, JIP staff and students within the same cluster. Teams should demonstrate that the feedback based on the rubric has been considered and incorporate into the work leading up to the mid-term review. However, the Midterm review uses the same criteria as the problem definition, re-evaluated at a deeper level. In addition to the review sessions, the JiP team monitors individual growth. The monitoring is done by reviewing a team blog, which shows the team's progress to peers in JIP and beyond. The team blog is meant to get feedback and keep track of the decisions students have made while working together concerning the project and team process. The individual process is realised by a 360-degree team feedback tool and team sessions to jointly evaluate the team process and improve the individual professional and collaborative attitude. (4 Team blogs/4 Personal Reflection blogs/ 4 time 360-degree team feedback) Summative assessment on module level Final review the presentation is behind closed doors and is open for academic staff, JIP staff and the company coach(es). Note that; the grade for the project report/presentation/prototype/model/design (80% of the final grade) will only be given if the two earlier reports (the two-pager on the problem statement review and the four-pager of the midterm review) are handed in as well. The items are assessed according to a rubric addressing interdisciplinary teamwork, scientific and technical rigour, responsible /ethical engineering, innovation and communication skills. The assessors (company, academic and Jip staff) will score the rubric independently, after which a final grade is calibrated and discussed with the entire committee. The Problem def/mid-term reviews and the growth that has occurred during the process are incorporated in the final grade. The committee consists of minimally, the company coach, one academic staff member with expertise on the topic, 1 (academic chair), 1 Jip staff. Often an additional expert from the field and academic staff member is added.	

	The minimum grade for the summative assessment is =/> 5.8. Process grade is 20% The team blog/personal blog is evaluated according to a rubric on interdisciplinary teamwork, professional development, academic attitude, personal leadership. The deliverables are subject to a knockout system. Meaning the grade is insufficient if not delivered. Students cannot finalise the project without handing in the obligatory work.

MSc 3 GM - Electives 10 EC

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Geomatics**

MSc 4 GM

GEO2020	Thesis	30
Course Coordinator	Dr. C. Garcia Sanchez	
Responsible for assignments	Dr. C. Garcia Sanchez	
Contact Hours / Week x/x/x/x	X/X/X/X	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	GEO2011	
Course Contents	The graduation work is the last step in the Geomatics education. It is completed in the form of an individual research project. GEO2020 follows GEO2011, and involves carrying out the research according to the plan of GEO2011. The topic of the scientific project can be related to any of the subjects offered by the Geomatics programme. The project must be supervised by one or more of the lecturers who participated in the Geomatics education. The project can be performed in cooperation with a company/institution, which provides a use case or a problem statement (but the work is always supervised by the scientific staff of the university). The graduation work deliverables are: (1) a scientific report (a thesis); and (2) an oral presentation.	
Study Goals	During the MSc thesis the student will show their knowledge, understanding and skills at an academic Masters level with respect to independently planning and executing a research project in the field of Geomatics. On completing the Graduation Project, the student is able to: 1. Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research; 2. Formulate a theoretical, numerical and/or experimental framework and delineate a research problem such that it can be solved; 3. Interpret obtained results in a critical manner; 4. Explain the work performed in a structured report that incorporates verification of methods and tools and is written in correct English; 5. Communicate the work performed in a structured way through an oral presentation to a wider audience; 6. Demonstrate capacity to manage the project, both technically and time-wise, considering resources and methodology;	
Education Method	individual work	
Practical Guide	All details are available at: https://3d.bk.tudelft.nl/courses/geo2020/ Brightspace is not used.	
Assessment	scientific report + presentation. These are evaluated by the graduation committee, guided by the geomatics graduation manual (i.e. the P4 and P5 assessments).	
Special Information	Geomatics students have the possibility to carry out their graduation research project in collaboration with a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found at https://3d.bk.tudelft.nl/courses/geo2020/company/. The agreements can be signed at the secretariat of Education and Student Affairs.	
Period of Education	All	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Geomatics**

GM Electives

ARFW0501	Applied Spatial Analysis for Sustainable Urban Development	5
Course Coordinator	C. Forgaci	
Course Coordinator	Dr. D. Cannatella	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Spatial analysis is a crucial part of projects carried out in urbanism and landscape architecture and a core application domain for the field of geomatics. After having encountered basic spatial analysis with short targeted workshops in the first three quarters of their MSc tracks, Urbanism and Landscape Architecture students following this intradisciplinary course work on an integrated, end-to-end analytical assignment, combining, advancing and extending their data-related skill- and tool-sets, while focusing on advanced workflows. The course offers the opportunity to MSc Urbanism Landscape Architecture, and Geomatics students to advance their technical skills required for scalable, automated and reproducible spatial analyses, practicing working in an intra-disciplinary setting with a real-world assignment. All students have the chance to learn skills from the other tracks and in the process strengthen their analytical skills for their upcoming thesis. The course will start with an introduction to multi-criteria decision analysis, typology construction and their corresponding workflows suitable for a sustainable spatial development challenge. The higher-level principles of scalability, automation and reproducibility, as well as their implementation through computational notebooks, are also introduced early in the course. The course emphasizes data visualization as a way to explore data iteratively, generate new knowledge and communicate project results. Students learn about different types of thematic analysis, such as spatial analysis, network analysis, and landscape analysis which they can integrate in their analytical workflows. To apply the workflows and analytical methods introduced throughout the quarter, students work in intra-disciplinary groups on a given location. The workflows presented in the course involve open-source tools using graphical user interfaces, such as QGIS, as well as programming in R and/or Python for spatial analysis. In the era of open data, the course emphasizes the importance of leveraging accessible and shared datasets to enhance the quality and depth of spatial analyses. In addition, the course underscores the significance of transparent and collaborative research practices by incorporating a dedicated segment on publishing datasets, workflows, and results. MSc Urbanism, Landscape Architecture and Geomatics students will be given the opportunity to contribute to the broader academic community through the dissemination of their work.	
Study Goals	After following this course, students will be able to: 1. Construct analytical workflows in the context of spatial design and planning. 2. Employ spatial data visualization both for exploratory data analysis and for the presentation of their final project results. 3. Conduct integrated spatial analysis to understand and describe the complexity of urban environments and to inform spatial design and planning decisions targeting their transformations. 4. Adopt open science practices, including FAIR data publication, reproducible workflows, and usage of open-source software. 5. Report on analytical findings as part of a multidisciplinary team.	
Education Method	The course has a blended learning set-up that makes use of online learning material developed or curated as part of the Spatial Data & GIS education platform of the Department of Urbanism. During contact hours, the focus will be on lectures introducing key concepts and hands-on workshops addressing the methodological and technical challenges of the workflows and methods introduced in the course. Self-study time will involve both individual and group work, with an emphasis on the latter. In addition to the online software tutorials, peer troubleshooting will be facilitated.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Students keep track of their progress and present their final products in a group report set up as a computational notebook in a version control system. This allows them to reflect on their process throughout the quarter and the instructors to assess individual contributions in detail. Peer feedback as an essential component of the course is employed in the midterm and the thematic workshops. A rubric will be used for grading. The rubric will be available in the course guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on one day part per week; on Tuesday or Friday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of	For this course the maximum number of participants is 30.	

participants	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

ARFW0503	**Urbidata: Responsible Design for Fair Data Cities**	**5**
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 7.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	In this course we explore the main principles of data ethics, and the relevance of applied ethics in the domain of the data-driven city. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence. Data, processed by AI and increasingly powerful computers, that promise mankind the opportunity to influence all aspects of life, especially in the urban setting. The concept of the 'smart city' assumes that big data leads to better (after all smart) solutions for planners, designers, industry and, at the end, citizens. Urbidata, the data-driven city of 2050, will be optimally planned, with smart mobility that allows for efficient and safe use of space and time, and citizens living in smart homes and working in smart offices that cater optimally their needs, based on their personal profiles. Does this technology driven perspective provide the answer how we want the life in the future city of Urbidata to look like? Should all developments of data-driven solutions (e.g. by architects, urban planners and geo-specialists) and their use be permissible in the light of the consequences? The choices we make today determine our world of tomorrow. Taking a critical, responsible design perspective it's clear that our data-driven future cannot be just a question of engineering and economic models. How about values as autonomy, privacy, transparency and consent? At first sight this interpretation of the city of tomorrow appears to be a question of policy. Policy that might lead to a normative (legal) framework setting standards. However in this course we focus on the preceding fundamental discussion. What are the ethical concepts and principles that guide us to determine what solutions are right or wrong? And next: how to apply data ethics in a given situation? This in order to make the future city of Urbidata a fair data city.	
Study Goals	After this course the student is able: - to explain the relevance of applied ethics in relation to the design and governance of the data driven city in general; - to identify concrete ethical issues within the field of data processing in a concrete situation; - to identify and apply the relevant ethical principles for this concrete situation; - to create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours). Self-study, Individual assignments (60 hours). Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The student will design an artefact and submit a written report, providing explanation of the artefact and containing a reflection. The artefact itself can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The student needs to give one presentation on an assigned topic during the course in order to pass the course. The course is concluded with an oral exam. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. A rubric will be used for grading. The rubric will be available on the course specific Brightspace page. Students should achieve a minimum grade of 5,75 to pass this course. Grades between 5,0 5,75 may be considered for a Repair. A Repair is provided when only one specific part (one Learning Objective) of the assignment is missing or insufficient. The student receives a well described written repair assignment. The assignment must be handed in by the student on Friday 12.00 h. in the first week of Summer, thus week 5.1. This will be mentioned in the written repair assignment. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 5,75 the student needs to re-enroll for the course.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on two day parts per week; on Tuesday and Friday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable	
Minimum number of participants	For this course the minimum number of participants is 15.	
Maximum number of participants	For this course the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5010	Research Assignment	2

Course Coordinator	Dr. C. Garcia Sanchez
Instructor	Dr. C. Garcia Sanchez
Responsible for assignments	Dr. C. Garcia Sanchez
Education Period	1 2 3 4
Start Education	1 2 3 4
Exam Period	none
Course Language	English
Course Contents	The course is meant as an introduction to one, or more, aspects of a research project. The student can do: 1. A literature study to get familiar with a related topic/domain that is not covered in the core programme of the MSc Geomatics. The knowledge learned can then be used as a start for a graduation project (GEO2010/2020), but it may also be unrelated to the thesis topic. 2. An implementation of concepts in a software prototype. This can be used for example to further explore one specific topic and to try to implement a prototype that will, for instance, allows us to process some datasets. This can be used to develop one's skills in a particular programming language. 3. A mix of 1 and 2 above. If a literature review is carried out, the main deliverable is a scientific report summarising the problem studied, the main findings, and potentially a discussion about future work related to the topic. If an implementation is carried out, then the student will deliver a software prototype, along with instructions about what it does, how to use it, test datasets, etc. Important note: If the research performed through GEO5010 is related to the MSc thesis, the work and output needs to be clearly distinguishable from the thesis itself, standing by its own merits. This work should neither be part of the thesis, but serve as an starting point that can be referred within the thesis bibliography.
Study Goals	After the course the student is able to: - formulate a clear problem definition with research questions; - independently carry out an assignment; - document the results of the assignment.
Education Method	There are no lectures scheduled for this course. Instead, an individual assignment is carried out. The assignment should be supervised by a staff of the MSc Geomatics programme at TU Delft, and has to be approved before the start by the course coordinator of GEO5010. The coordinator should receive a copy of the proposed assignment before the actual start of the work, and of the final deliverable at the end of the assignment. To start the project, a half-page summary containing the following is required: - name student - title of project - main objectives and deliverables - planned timeframe
Assessment	The marking of an assessment is made by the supervisor of the project and the coordinator of the course (in order to guarantee uniformity of marking in relation to the level of difficulty of different assignments). There is in theory no time limit for an assignment, that is a student can start for instance during the summer and work part-time on the project for a few months. An assignment can be handed-in at any time during the year.
Special Information	The maximum marking period is 10 work days.
Remarks	Brightspace is not used, all the information is there: https://3d.bk.tudelft.nl/courses/geo5010/
Period of Education	no restriction, a project can start/finish at any time of the year
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	Written exam (3 hours) and formative assignments. The exam result will determine the final score, but assignments need to be sufficient (>=5,75). The main purpose of the assignments is to support the understanding of the theory via hands-on experience. The exam is the only summative assessment and determines your grade for 100%. However, students need to pass at least three of the four formative assignments in order to pass the course. In case of non-passed assignments, in week 9 of the course, students have the opportunity to repair these non-passed assignments.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Period of Education	Quarter 3.	
Concept Schedule	Tuesday 3e/4e hour + Friday 1e/2e hour (all with reservations)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	The summative assessment consists of two graded assignments, and one final oral exam. The final grade is the average using these weights: 35% Assignment 1, 35% Assignment 2, 30% final oral exam. The final grade must be >= 5.75 to pass the course. Following conditions apply: - The average grade of both assignments must be >= 5.75 to pass. If not, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be >= 5.75 to pass. If not, there is a retake oral exam at the end of Q2. - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	Dr. C. Garcia Sanchez	
Responsible for assignments	Dr. C. Garcia Sanchez	
Contact Hours / Week x/x/x/x	4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Monday, Tuesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5016	Geomatics in practice	10
Course Coordinator	F.M. Welle Donker	
Responsible for assignments	F.M. Welle Donker	
Contact Hours / Week x/x/x/x	The number of contact hours per week varies per internship.	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	You should have completed at least 30 ECTS of core Geomatics Programme MSc 1 and MSc 2. This is a non-negotiable condition.	
Course Contents	The Geomatics in Practice Elective provides an opportunity to apply knowledge and skills obtained in the first year of Geomatics for the Built Environment in practice. The Internship must have an academic character.	
Course Contents Continuation	There is no formal start date or end date, you can start any time provided you have completed at least 30 ECTS of Geomatics core subjects. The internship should last at least 280 hours, i.e., seven weeks on a full-time basis, or an equivalent period on a part-time basis, e.g., 14 weeks @ 20 hours/week. BEFORE YOU START: When you have found an internship at a suitable academic level, you should submit an Internship Proposal form (maximum 800 words) via Brightspace, stating: 1. Your name, study number, email address, and completed geomatics core courses 2. Short description of the company (max 1 paragraph) 3. Name, affiliation, and qualifications of the supervisor at the company/organization (your Workplace Supervisor) 4. Name, and affiliation of proposed geomatics tutor (your Geomatics Supervisor) 5. Title / Topic of the work you will carry out, objected achievements, and deliverables to the organization 6. Motivation for why you want to carry out this work and indication of how it will contribute to your geomatics knowledge and skills 7. Indication of the academic level of the work you will carry out (e.g., design, data integration, modeling, assessment) and whether individual or as part of a team 8. Location of where you will be doing the activities (if partially working from home, please indicate the percentage of time working on location and working from home) 9. Time planning: start data + finish date + no. of hours per week Please involve your Geomatics Supervisor when you draft your internship proposal. The Geomatics Supervisor may be any person connected to the Geomatics programme with sufficient knowledge of your chosen topic. When you submit your Internship Proposal via Brightspace, please inform the GEO5016 Coordinator by e-mail of your submission. The GEO5016 Coordinator will assess your Internship Proposal for suitability for the Geomatics programme. A GO will be granted when the internship Coordinator is satisfied that the proposed topic and activities are suitable for the Geomatics Programme and at MSc level, i.e., there has to be an academic challenge. This means that you have to work towards tackling a problem, working with complex and heterogeneous data, or developing a model or something similar, either individually or as part of a team. So, just carrying out activities according to fixed protocols without a personal contribution is not sufficient. Without a GO, you are not allowed to start this internship. Do not forget to enroll in GEO5016 Geomatics in Practice, Aanvangsblok: JAAR. WHEN YOU RECEIVE A GO: Submit a UNL Internship Agreement, signed by you and your Workplace Supervisor, to J.vanOs@tudelft.nl with a cc to the GEO5016 Coordinator. The Internship Agreement form can be downloaded in Dutch and English from https://www.universiteitenvannederland.nl/onderwerpen/onderwijs/gemeenschappelijke-stageovereenkomst-universiteiten. DURING YOUR INTERNSHIP: Carry out your internship activities and absorb as much practical experience as you can! AFTER YOUR INTERNSHIP: Submit your internship assessment form to your workplace supervisor for assessment of your professional skills no later than 1 week after completion of your internship (30% of the final mark) Submit your Internship Report and internship assessment form to your Geomatics Supervisor with a cc to the GEO5016 Coordinator no later than 4 weeks after completion of your internship. Make an appointment with your Geomatics Supervisor to discuss your report and to gain an assessment mark (70% of the final mark).	
Study Goals	To experience working in practice, in a company, at a (local) governmental institute or a research institute. To apply knowledge from the Geomatics Programme in a practical project.	
Education Method	Individual work: Internship Report (max 5,000 words) according to the template provided via Brightspace.	
Practical Guide	You can start any time you want, there is no formal start date or end date, provided you have completed at least 30 ECTS of Geomatics core subjects. This is a non-negotiable condition. The internship should last at least 280 hours, i.e., seven weeks on full-time basis, or an equivalent period on part-time basis, e.g., 14 weeks @ 20 hours/week. It is NOT possible to apply for this course or to obtain study points based on an internship in the past. The internship must be approved on forehand. The TU Delft internship contract is required for the project. BEFORE YOU START: Find an internship at a suitable academic level. Find a Geomatics Supervisor. Submit an Internship Proposal form to the GEO5016 Coordinator to receive a GO. Without a GO, you are not allowed to start this internship or receive study points for GEO5016. Enroll in GEO5016 Geomatics in Practice, Aanvangsblok: JAAR. WHEN YOU RECEIVE A GO: Submit a UNL Internship Agreement, signed by you and your Workplace Supervisor, to J.vanOs@tudelft.nl with a cc to the GEO5016 Coordinator. The Internship Agreement form can be downloaded in Dutch and English from https://www.universiteitenvannederland.nl/onderwerpen/onderwijs/gemeenschappelijke-stageovereenkomst-universiteiten. Only	

	the UNL Internship Agreement forms are accepted, and not a proprietary contract issued by the internship organization. DURING YOUR INTERNSHIP: Carry out your internship activities and absorb as much practical experience as you can! AFTER YOUR INTERNSHIP: Submit your internship assessment form to your workplace supervisor for assessment of your professional skills no later than 1 week after completion of your internship (30% of the final mark) Submit your Internship Report and internship assessment form to your Geomatics Supervisor with a cc to the GEO5016 Coordinator no later than 4 weeks after completion of your internship. Make an appointment with your Geomatics Supervisor to discuss your report and to gain an assessment mark (70% of the final mark).
Prerequisites	You should have completed at least 30 ECTS of core Geomatics Programme MSc 1 and MSc 2. This is a non-negotiable condition
Assessment	The final assessment will be based on the Professional Skills (30%) and the Internship Report (template)(70%). The Internship Report should be written in English. The Professional Skills are based on 8 criteria (rubric): 1.1 Creative thinking & showing initiative 1.2 Productivity 1.3 Independence 1.4 Motivation & enthusiasm, commitment & perseverance 1.5 Critical attitude 1.6 Handling supervisor's comments & development skills 1.7 Interpersonal / insight in functioning of workplace organisation 1.8 Time management The Internship Report is an Academic Paper which will be assessed on 8 criteria (rubric): 2.1 Context of research / internship topic & problem statement 2.2 Formulation goals, tasks & framework of internship topic 2.3 Theoretical underpinning, use of literature 2.4 Use of methods & processing / analysing data 2.5 Reflection on results 2.6 Conclusions and discussion 2.7 Fluency of language and writing skills 2.8 Personal reflection on internship The final delivery is an Internship Report that consists of the following components: - Abstract a summary of the report, max 200 words - Introduction the background and context of the topic including a problem definition: Why this project? - Objectives the objective of internship project and the research questions: What will the project deliver? What question will be answered? - Methodology the proposed approach and method: How will you answer the research question? Which components, steps or working packages do you distinguish? - Results What did you do and what are the outcomes of your project? - Analysis and discussion What do the outcomes mean? Are there any gaps? - Reflection what are the proposed learning objectives? How far did you meet those or extend them (performance of the project)? Why did you divert from the original objectives? Did the project connect to your prior knowledge is there a gap? Which courses did the project directly connect to (motivate)? A Retake of GEO5016 is possible if any of the assessments of Part A (Professional Skills) or Part B (Academic Skills Internship Report) are 5.0 or lower, i.e., if the assessment of either the workplace supervisor (30% of final mark) or of the Geomatics supervisor (70% of final mark) is 5.0 or lower. A Retake means that the student must do a new internship. As this is a new chance, the maximum mark of the Retake Internship will be 10.0. A Repair is only possible if the assessment of Part B (Academic Skills Internship Report) is between 5.0 and 5.74 and the assessment of Part A (Professional Skills) is 5.75 or higher. A Repair is a rewrite of the Internship Report, and the maximum mark will be 6.0 out of 10. There is no option for a Repair if the assessment of Part A (Professional Skills) by the Workplace Supervisor is between 5.0 and 5.74. *It is NOT possible to apply for this course or obtain study points based on an internship in the past. The internship must be approved on forehand. The UNL internship agreement is required for the project.
Enrolment / Application	BIS: only for Geomatics students
Special Information	Definition Internship Coordinator = coordinator of this course Workplace Supervisor = supervisor at the organization supplying the internship. He/she should have a geomatics background and a MSc degree. Geomatics Supervisor = supervisor from TU Delft Geomatics MSc Programme and assigned to the internship based on the proposal. Procedure: Step 1: submit a proposal to the coordinator (form). The proposal will be assessed by the coordinator and proposed academic supervisor. Step 2: if the proposal receives a GO, the UNL Internship Agreement should be signed by the student, the internship organization, and TU Delft. Step 3: The internship is carried out* Step 4: An Internship Report is submitted to the Internship coordinator. The internship organization may request a separate project report or presentation. The proposal Before starting the project a proposal must be submitted to the Internship Coordinator. The proposal is a maximum of 800 words / 2 A4 and contains the following elements: 1. Where will the internship be carried out? Name and address of the company, short description of the company (1 paragraph). 2. Who will be the internship supervisor? Name, affiliation, and qualifications. This supervisor needs to have a background in a geomatics-related study. 3. Who will be the proposed academic supervisor? Name, affiliation. 4. Field: what field or theme is the internship covering? What is the context of the internship? What are the (academic or practical) challenges? Why this internship? 5. What will be the proposed assignment or topic of the internship? For the proposal a draft/sketch is fine. A final assignment will be defined in week 1 of the internship. 6. Planning and duration: start date, end date, duty/commitment (fte), modus of attendance (in percentages), scheduled holidays.

	The minimum duration for the internship is 140 hours, excluding public holidays. 7. Learning objective(s) for practice: at least one personal learning objective; definition of start level and proposed end level. 8. Learning objective(s) for geomatics: at least one academic learning objective; definition of start level and proposed end level. Evaluation of the proposal is based on: - Relation to Geomatics of the topic and assignment - The academic level (MSc) of the proposal - Practical work challenges Rubric / Acknowledgement For the assessment, the Rubric for the Assessment GEO5016 is used.
Period of Education	not applicable
Concept Schedule	not applicable
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Instructor	Dr. L. Nan	
Instructor	Ir. S. Du	
Instructor	N. Ibrahimli	
Responsible for assignments	Dr. L. Nan	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-from solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropogation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on	

	the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. To pass the course, the following applies: 1. a total average of 5.75 or above is required to pass the course, and 2. all assignments and the final exam must be graded greater or equal to 5.5. Repair and retake options: any component of the assessment graded 5.75 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	**2024/2025**
Organization	**Architecture and the Built Environment**
Education	**Master Geomatics**

Electives, courses listed below are recommended to be chosen as electives

AESM5110C	Aerosol and Cloud Microphysics	5
Responsible Instructor	Dr. F. Glassmeier	
Instructor	I.M. Steinke	
Co-responsible for assignments	Z.C. Rowland	
Contact Hours / Week x/x/x/x	4 hours lectures & 2 hours practical	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	.	
Study Goals	After completing this module, you will be able to estimate direct effects of - uncertainties related to aerosol- and cloud microphysical parameterization for weather prediction and climate projection; - inadvertent anthropogenic aerosol emissions and their reductions on weather and climate; - targeted aerosol injections as part of weather modification and climate-engineering proposals on weather and climate; - precipitation on particulate air pollution and to evaluate research literature on detailed quantifications of these effects. You will specifically be able to: - Mathematically model important source, sink and transformation processes of liquid and frozen cloud water and atmospheric aerosol particles. - Design, perform and interpret numerical experiments and predictions with an aerosol-cloud parcel model. - Summarize and evaluate the limitations, challenges, and uncertainties of predicting aerosol and cloud microphysics and aerosol cloud interactions.	
Education Method	- lectures (semi-weekly) - tutorial class (weekly) - homework assignments (pen and paper calculations, numerical simulations, weekly during first part of the course) - final project (paper discussion + analytical calculations or numerical simulations to evaluate its limitations and uncertainties)	
Assessment	Assessment (Formative): - in-class polling - homework correction/discussion - pre-report presentation of final project Assessment (summative) - Written report that documents the final project (100%) - A minimum of 5.8 is required to pass this course.	
Expected prior Knowledge	Essential: introductory physics and mathematics, esp. thermodynamics and simple differential equations Recommended: introductory Earth and/or environmental science; students from Applied Physics, Aerospace Engineering or similar are welcome (please have a look at the suggested textbooks to judge the match with your background)	
Academic Skills	-	
Literature & Study Materials	- Lamb & Verlinde: Physics and Chemistry of Clouds (selected sections, available as e-book at TUD library) - Lohmann, Lüönd, Mahrt: An introduction to clouds from the microscale to climate (selected sections, available as e-book at TUD library) - comprehensive aerosol-cloud parcel model with ipython notebook interface - selected research articles	
Judgement	-	
Permitted Materials during Exam	Not applicable	
Collegerama	No	

AESM5120C	Data Assimilation for the Geosciences	5
Responsible Instructor	Prof.dr.ir. F.C. Vossepoel	
Instructor	Dr.sc.nat M. Ramgraber	
Instructor	Prof.dr.ir. A.W. Heemink	
Instructor	Prof.dr.ir. M. Verlaan	
Instructor	Dr.ir. R.W. Hut	
Contact Hours / Week x/x/x/x	2 hours lectures & 3 hours practicals	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course contains the following topics: Bayesian formulation of data assimilation Variational methods (4DVar) Ensemble smoothers (EnRML, ESMDA) Ensemble Kalman Filter (EnKF) Particle Filter Appplications of data assimilation in oceanography, meteorology, hydrology, seismology, reservoir engineering and/or geotechnics; Open and FAIR science, open interfaces	
Study Goals	1. To explain the Bayesian principles of data assimilation 2. To relate selected data assimilation methods to these principles (Strong-constraint 4Dvar/Weak-constraint 4Dvar/EnRML, ESMDA/EnKF/Particle Filter) 3. To evaluate existing data-assimilation methods and select the most appropriate method for a given data-assimilation problem 4. To design a data-assimilation workflow that connects a selected model of a particular earth system (oceanographic, meteorologic, hydrologic, reservoir engineering, seismology, geotechnical) to a data assimilation method, or of another estimation problem (traffic, medical sciences, finance) 5. To apply open interfaces 6. To apply a data-assimilation method to a proposed data-assimilation problem	
Education Method	Lectures (2 hours per week) and practical assignments (3 hours per week), self-study	
Course Relations	Can be combined with the Applied Mathematics course Data Assimilation WI4475 on algorithms for data assimilation. The course also links to the cross-over course Advanced Topics in Probabilistic and Statistics for Engineeris in the second quarter of year 2 of the MSc programmes.	
Assessment	Assessment (Formative) In-class assignments Assessment (summative) Report of an individual assignment (application of data assimilation in a problem selected by the student) (100%) A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? AES, EE, CT	
Expected prior Knowledge	Knowledge on statistics and partial differential equations as taught in Modelling, Uncertainty and Data for Engineers (CEGM1000)	
Academic Skills	-	
Literature & Study Materials	Data Assimilation Fundamentals, A Unified Formulation of the State and Parameter Estimation Problem, by G. Evensen, F.C. Vossepoel and P.J. van Leeuwen, open access: https://link.springer.com/book/10.1007/978-3-030-96709-3 and selected publications (online)	
Judgement	-	
Permitted Materials during Exam	-	
Collegerama	No	

AESM5210C	Climate Remote Sensing	5
Responsible Instructor	Dr. P.G. Ditmar	
Instructor	Prof. Dr.-Ing. habil R. Klees	
Instructor	Dr.ir. B. Wouters	
Instructor	Dr.ir. D.C. Slobbe	
Contact Hours / Week x/x/x/x	6 h / week (weeks 1-6); 2 h / week (weeks 7-10);	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The module is set up to meet the growing public concern about climate change, sea level rise, and potential shortage of freshwater resources, which are consumed with an increasing rate. To address these concerns, it is vital to measure, unravel, forecast, and, if possible, control various climate- and water-related processes. In this context, the data delivered by remote sensing techniques play a central role. In this module, students will learn how to process, invert, and combine data, including those of different spatial/temporal resolutions. Moreover, they will learn how a proper data weighting can be implemented in order to optimize results of data combination. Using data provided by satellite altimetry and satellite gravimetry missions as an example, students will be able to produce state-of-the-art estimates of large-scale mass change in various Earth system compartments, such as the melting of glaciers and ice sheets, climate-driven variability in river basin water storage, and depletion of groundwater stocks.	
Study Goals	After completing this module, students will be able to: 1. Design a scheme for level-2 satellite gravimetry and satellite altimetry data processing, taking into account data uncertainties 2. Design efficient numerical schemes for a stand-alone and joint data inversion that take into account stochastic models of data errors, possible inconsistencies between datasets, as well as available prior information 3. Apply Kalman filtering to assimilate data into a simple numerical model and assess the added value of data assimilation 4. Apply stand-alone and joint inversion schemes to quantify and separate processes at different time scales in the context of cryosphere, ocean, or hydrology 5. Summarize and defend the findings of a conducted study of selected Earth system processes	
Education Method	Lectures, in-class assignments, individual weekly homework assignments, individual integrated project at the end (combined with an oral presentation and a written report)	
Course Relations	The students will benefit from basic knowledge of potential fields and spherical harmonics, which are taught, e.g., in course AESM2001 (Physical principles of Earth System Observation)	
Reader		
Assessment	Assessment (formative): Guiding in-class assignments (feedback in real time) Providing feedback to (timely) completed weekly homework assignments Assessment (summative): Portfolio composed of: (100%) Completed homework assignments Oral presentation Report on the integrated project (no exam at the end) A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? MSc AES, MSc CE, MSc EE	
Expected prior Knowledge	Basic knowledge of Python language Parameter estimation in linear models Basics of statistical theory Spherical harmonics (recommended)	
Academic Skills	Conceptual thinking, critical thinking, problem solving and creativity, communicating and presentation skills	
Literature & Study Materials	Lecture slides, lecture notes, material for further reading (e.g. scientific articles)	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	N/A	
Collegerama	No	

AESM5220C	Microwave Remote Sensing of the Earths Surface	5
Responsible Instructor	Prof.dr.ir. S.C. Steele-Dunne	
Instructor	Dr.ir. B. Wouters	
Instructor	Prof.dr.ir. F. Lopez Dekker	
Contact Hours / Week x/x/x/x	x/x/x/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	By the end of this course, students will be able to work with, and perform research based on real or synthetic data from current and imminent microwave missions (e.g. Sentinel-1, ROSE-L, CIMR, Harmony, Metop ASCAT/SG-SCA) to study processes relevant in land, ocean and cryosphere applications, in particular processes related to surface-atmosphere interactions. Microwave dielectric properties of natural materials Microwave remote sensing of soil and vegetation. The relation between surface soil moisture, root zone soil moisture and vegetation water content variations and their role in land-atmosphere exchanges of water, energy and carbon. Modelling the influence of dielectric properties and geometry on emission and scattering from vegetated surfaces. Soil moisture estimation from passive and active microwave remote sensing. Monitoring biomass and vegetation water status using passive and active microwave remote sensing. Land-atmosphere interactions over ice/snow The relation between physical snow/firn/ice properties and land/atmosphere interactions for ice The relation between snow/firn/ice properties and the EM properties in the MW region Radiative transfer models to translate snow/firn/ice properties into remote sensing signals for both passive/active MW RS sensors Retrieval of snow/firn/ice properties from MW RS data Ocean The relation between wind, marine boundary layer conditions, and directional surface wave spectra. Theoretical models relating wave-spectra and the resulting directional roughness to the radar scattering intensities and microwave emissivity. Empirical Geophysical Model Functions (GMF) relating surface winds and/or wind stress to radar scattering and Doppler Retrieval of surface wind and surface wave information using data from radar-scatter meters, Synthetic Aperture Radars, and microwave radiometers.	
Study Goals	After completing this module, students will be able to: 1. Explain and describe the relations between surface properties and processes, and observations that can be obtained using microwave remote sensing. 2. Apply state-of-the-art models and retrieval techniques to simulate microwave observables and retrieve states of interest in land, ocean and cryosphere applications . 3. Compare different forward modelling or retrieval techniques to estimate variables of interest at the surface 4. Select and defend the choice of a product/technique/model to capture a process relevant in land, ocean and cryosphere applications.	
Education Method	Video lectures, in-class lectures & discussion	
Assessment	Assessment (Formative) Weekly assignments Assessment (summative) An assignment portfolio, with a final oral defense (100%) A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? AES Students from other faculties or universities with an equivalent background can follow this course with approval of the lecturer.	
Expected prior Knowledge	Program Core PPESO or equivalent (check with lecturer)	
Academic Skills	-	

Literature & Study Materials	Lecture notes, online material for further reading (e.g. scientific articles), extracts of text books e.g. Ulaby, Rees, materials provided in class
Judgement	-
Permitted Materials during Exam	-
Collegerama	No

AESM5230C	Coastal Remote Sensing	5
Responsible Instructor	Dr. R.C. Lindenbergh	
Instructor	Dr.ing. J.A. Antolínez	
Instructor	Dr. A. Amiri Simkooei	
Instructor	Dr.ir. F.J. van Leijen	
Instructor	Dr.ir. M.A. de Schipper	
Instructor	Dr.ir. A.P. Luijendijk	
Contact Hours / Week x/x/x/x	2 hours lectures & 2 hours practical	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The Coastal Remote Sensing module will provide students with theoretical background on both remote sensing and coastal processes. In this module an overview of coastal processes will be given that are relevant in an applied context, from the coastal shelf to the dunes. It will be discussed at which spatio-temporal scales coastal processes occur, and how these scales relate to remote sensing techniques, from in situ onshore and offshore techniques to satellite systems. An overview will be given of remote sensing techniques for measuring bathymetry and topography, including echo-sounding, laser scanning and photogrammetry, but also the role of spectral techniques will be discussed, to e.g., assess sediment load, sea water temperature and dune vegetation properties. Finally, processing techniques to extract information from data, including data fusion techniques, time series analysis, uncertainty assessment and relevant machine earning methodology will be covered.	
Study Goals	On completion of this module, the student will be able to: Explain which remote sensing techniques are suitable to assess coastal processes in terms of the spatio-temporal scales at which they occur. Develop a workflow to answer a research question on coastal processes using a suitable combination of remote sensing techniques. Extract information from coastal remote sensing data and evaluate the results to answer a research question Associate an error budget to different kind of coastal remote sensing data products and apply this error budget to assess the uncertainty of work-flow outcomes. Defend the results of an implemented workflow.	
Education Method	Lectures. Lectures will be closely supported by dedicated lecture notes, consisting of a mix of existing and newly developed material. In addition, relevant scientific literature will be shared via the lecture notes and Brightspace. During the lectures, students will be engaged by questions (partially hardcoded in the slides, which means that questions are formulated as questions on the slides) and exercises. Instructional videos. Will be shared to shortly explain different concepts on remote sensing and coastal processes. This is also a way to make material available that goes beyond a common core.	
Course Relations	This Module is intended to be closely related to Module CIEM4304 Hydraulics Fieldwork as offered in Q5 within the Track Hydraulic Engineering (Module Manager: Matthieu de Schipper)	
Assessment	Assessment (Formative) Feedback will be organized around informal quizzes, during each lecture Students will be asked each lecture to write down 2 topics they liked and 2 topics that were not sufficiently clear. Assessment (summative) Assignment (100%). Assignment based. In assignments, students will obtain hands-on experience with coastal remote sensing data related to specific coastal processes. Assignments will be organised in groups of ~3 students A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? Notably: Discipline Earth Observation within the programme Applied Earth Sciences Track Hydraulic Engineering within the programme Civil Engineering	
Expected prior Knowledge	Modelling, Uncertainty and Data for Engineeers (CEGM1000) or equivalent Python programming Below courses/modules/units are not expected prior knowledge, but are related: MSc CE, Track Hydraulic Engineering (HE):	

	A-Module, Hydraulic Engineering Unit: Coastal and Estuarine Systems B Module, Coastal Engineering: Unit: Advanced Coastal and Estuarine Systems, Unit: Coastal and Estuarine Modelling, Loads & Protection MSc AES, Discipline Earth Observation Disc. Core: Earth Observation Technologies A-Modules: Unit, Statistical geo-data analysis Unit, Geo-informatics B-Modules: Unit: GNSS Unit: (In)SAR Unit: Optical Remote Sensing Unit: (GR + EO): Geostatistical Data Analysis Unit (Track: Climate &Weather): Sea level change and floods
Academic Skills	Teamwork, independency, problem solving, creativity, flexibility, time management, communication skills, critical thinking, analytical skills
Literature & Study Materials	Lecture notes Books, with a preference for books freely available to students as eBook. Relevant titles include: Tiberius, C. C. J. M., van der Marel, H., Reudink, R. H. C., & van Leijen, F. J. (2021). Surveying and Mapping, https://doi.org/10.5074/T.2021.007 Bosboom + Stive, specific chapters from Coastal Dynamics. https://doi.org/10.5074/T.2021.001 Rees, W. G. (2013). Physical principles of remote sensing. Cambridge university press. Vosselman, G., & Maas, H. G. (2010). Airborne and terrestrial laser scanning. CRC press. Lurton, X. (2002). An introduction to underwater acoustics: principles and applications. Springer Science & Business Media. Short instruction videos
Judgement	-
Permitted Materials during Exam	N/A
Collegerama	No

AESM5240C	Applied Space Geodesy	5
Responsible Instructor	Prof.dr.ir. R.F. Hanssen	
Contact Hours / Week x/x/x/x	2 hours lectures & 2 hours project	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Satellite Earth Observation and Space Geodesy is an increasingly fertile field, with exciting technology that leads to a wide range of applications. Pioneers in the field are often young entrepreneurial start-ups that provide solutions that are directly applicable in society. Examples include, e.g., Planet, Maxar, Iceye, and Capella. What makes those initiatives so successful, and how can we develop new opportunities and challenges that have high impact? In this course you will create your own start-up based on Satellite Earth Observation. We discuss, e.g., core space technologies, value proposition design and strategies, data processing, funding, IP, scalability, valuation, and collaborate as a team to develop a new Earth Observation product or service. We evaluate current European policies to stimulate value creation from Earth Observation, and review legal restrictions. We will use guest lectures from space industry leaders and entrepreneurs as well as site visits to learn about the challenges and pitfalls. Our information products will be used to respond to scientific or societal challenges. Examples include information products for the Dutch Urban Search and Rescue (USAR) teams, requiring emergency response information, or energy sustainability for solar farms.	
Study Goals	After completing this module, students will be able to: Evaluate the space technologies available for value creation Develop an idea into an entrepreneurial business model and a new Earth Observation product or service Create a start-up up to a Minimal Viable Product	
Education Method	In-class lectures, video lectures, site visits, and discussion	
Assessment	Assessment (Formative) Weekly assignments Assessment (summative) Final written report and presentation (100%) A minimum of 5.8 per exam/assessment is required to pass this course.	
Remarks	Which programmes/tracks can follow this module? Aerospace Engineering, Computer Science, Geomatics, Civil Engineering, Applied Earth Science	
Expected prior Knowledge	-	
Academic Skills	-	
Literature & Study Materials	Lecture notes, online material for further reading (e.g. scientific and professional articles), extracts of text books, materials provided in class	
Judgement	-	
Permitted Materials during Exam	N/A	
Collegerama	No	

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Pereira Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Pereira Roders	
Co-responsible for assignments	Y. Zhang	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Students work in groups on a selected case study. Each group is expected to present a coherent research result on the impact of architectural redesign. The research will be divided among group members, and each student chooses his/her domain in the case study. The five domains are: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (i.e. social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (i.e. attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Course Contents Continuation	Lectures: Students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Seminars: Two seminars are organized for the students, including a series of lectures and discussion. Seminar on research invites researchers to showcase state-of-the-art research related to the domains. Seminar on Practice invites practitioners from heritage, architecture, consultancy, as well as local government to share their experiences in redesign projects. Tutorials: Two tutorials are offered to monitor and give feedback on the progress of their research. The students will meet per project, as well as per domain to discuss their research with the tutors. Self-study: The students are expected to be proactive in doing the research and preparing the assignments. Field work include but not limited to: archival research, conducting interviews with stakeholders, building survey and document analysis, etc. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their research results (codes) in pecha-kucha format. The code of the architect will be revealed and compared to the one defined by the students. By the end of the course, the students will compile the research in a written report.	
Study Goals	Master critical thinking in architectural redesign, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of the architectural redesign. In terms of knowledge, the students will: Assess a selected domain, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework In terms of skills, the students will: Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as explain their interrelations. Produce datasets, documentaries and argue in discussions with team members and architects, using an appropriate professional scientific language In terms of attitude, the students will: Develop an investigative attitude towards the nature and impact of architectural redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain.	
Education Method	Lectures (including Seminars): 12 hours Tutorials/Presentations: 16 hours Group field work: 4 hours Independent study (including additional field work): 108 hours (77%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Book chapters, journal articles and other lecture materials.	
Assessment	Dataset 40% Pechakucha 20% Final Report 30% Attitude 10%	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5

Course Coordinator	Dr. B. Lubelli
Instructor	T.P. Bennebroek
Instructor	Dr. B. Lubelli
Instructor	S. Naldini
Responsible for assignments	Dr. B. Lubelli
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)
Education Period	3
Start Education	3
Exam Period	3
Course Language	English
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)
Study Goals	At the end of the course, the student : has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. is able to document study results visually, in written text and verbally using appropriate technical language
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date for submission of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report. The maximum marking period is 10 working days.
Period of Education	Q3
Concept Schedule	Wednesday afternoon
Minimum number of participants	15
Maximum number of participants	60
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0916	Beyond 3D Computer Visualisation	5
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Responsible for assignments	P. de Ruiter	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practiced.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	A (high end) gaming laptop is required (minimum specification TU Delft student laptop).	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Enrolment / Application	Students are required to register for courses and exams in both the Bachelor's and Master's programmes. For more information, see: https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/enrolment-education-exams Additional enrolment information for non-BK students: Course enrolment via Brightspace only is not considered a valid registration Enrolment in the BK-enrolment system BIS (https://bis.bk.tudelft.nl/) does not guarantee placement; there may be a maximum number of non-BK participants for this course (see below) The course can be taken only after confirmation of placement in BIS by BK-enrolment	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Lectures on Monday, 10:45-12:30 Workshops on Wednesday or Tuesday, 8:45-12:30	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

CESE4055	Ad hoc and Sensor Networks	5
Responsible Instructor	Dr. R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Wireless communications and networking Computer communication principles, Layering principle of Computer Networks. Digital communication.	
Course Contents	IMPORTANT NOTICE ------------------------- Please note that the prevailing conditions of the COVID19 may force us to modify the study guide information. It may change depending on the developments around the coronavirus. We may have to make changes to the teaching methodology, assessment, practical work, assignment and group activities. This will be instructed by the faculty/university management based on the orders of the government from time to time to protect the faculty and students. The above applies to all the fields in this coursebase for this course. The course will be offered in PHYSICAL format. -- Ad-hoc networks are formed in situations where mobile computing devices require networking applications when a fixed network infrastructure is not available or not preferred to be used. In such cases, mobile devices may possibly set up an ad hoc network themselves. Ad-hoc networks are decentralized, self-organizing networks and are capable of forming a communication network without relying on any fixed infrastructure. Ad-hoc networks form a relatively new field of research. In this lecture, besides general introduction to ad-hoc networks and their applications, we will focus on state-of-the-art methods and technologies for forming an ad-hoc network and maintaining its stability despite the dynamics of the network. The contents of the course are as follows: Positioning and applications (Chapter 1, 2 & 3 of the textbook, these topics are basics & pre-requisites; And Chapter 5) o Definition of ad-hoc networks o Comparison with infrastructure based systems o Typical applications o Advantages and challenges o Radio technologies for ad-hoc networks o Wi-Fi, Zigbee, Bluetooth Modelling ad-hoc networks o Propagation models o Topology models based on graph theory o Degree and hopcount o Connectivity theorems MAC protocols for ad-hoc networks (Chapter 6, 10 of the textbook) o Introduction to MAC protocols o Issues and design goals o Classification o Directional, muli-channel MAC protocols o Energy efficiency in MAC protocols o Quality of service Self organisation and Routing (Chapter 7, 8, 11 of the textbook) o Flooding o Node discovery, neighbour discovery o Route establishment o Topology maintenance, localisation o Proactive, reactive and hybrid routing o Typical protocols o Energy efficiency in routing o Broadcast and multicast o Effects of mobility on connectivity and capacity o Effect of nodes joining and leaving the network Advanced issues in ad hoc networks o Wireless sensor networks (Chapter 12 of the textbook and papers) o Cooperation (Reference papers) o Simulating ad hoc networks as part of project (optional: ns3, OMNET, OPNET) o Energy Harvesting Project presentations by students	
Study Goals	By the end of this course students should be able to: - Model the ad-hoc networks using Graphs. - Describe the working principles of medium access control protocols for ad-hoc networks - Explain the working principles, advantages and disadvantages of different classes of routing protocols for ad-hoc networks - Choose various components to form a coherent ad hoc networking architecture - Develop a simulator to evaluate the MAC and routing protocols for ad hoc networks - Assess the suitability of ad-hoc networks for different communication needs and scenarios	
Education Method	The course will be taught in lecture form. The presence of students at all lectures is required for optimum result. Students are required to participate actively in various forms of activities and peer-learning. New forms of teaching aids are used.	
Literature and Study Materials	1. Textbook: Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. 2. Lecture notes consist of slides presented at the lectures (Slides are only teaching aids and are not substitutes for textbooks,	

	research papers, etc). 3. Some recent journal papers 4. Optional Reference Books 4.1. Distributed Algorithms, Nancy A. Lynch, Morgan Kaufmann, 1996 (for networking algorithms) 4.2. Ad Hoc Mobile Wireless Networks, Principles, Protocols and Applications by Subir Kumar Sarkar, C Puttamadappa, and T. G Basavaraju, Auerbach Publications, 2008. This book is available online in the library. 4.3. Wireless Ad Hoc and Sensor Networks, A Cross-Layer Design Perspective by Jurdak, Raja, Springer, 2007. This book is available online in the library. 4.4. Ad-hoc Networks: Fundamental Properties and Network Topologies, by Ramin Hekmat, Springer. 5. OPNET/ns-2 web pages, tutorials and video lectures
Books	Ad Hoc Wireless Networks, Architectures and Protocols by C. Siva Ram Murthy and B.S.Manoj, Prentice-Hall Pearson, 2004. However, I also use other materials from Internet and other books listed above.
Assessment	1. There will be totally three written/oral tests/examinations for this course (including the final exam). 2. The students will carry out a project in a group and submit a short report. 3. Participation in off-track discussions on Facebook/Brightspace/FeedbackFruits and wikis. Final score is based on marks obtained during tests, project, assignment (in groups) and bonus marks. All the details will be given in the first class. Breakup: 2 Tests + Final Exam = 55% Project 40% Self assessment + Reflection 3% Activities on Feedback Fruits or any Online platform 2% Resit: written/oral exam [+ carry over from projects]. == Changes due to COVID19 in 2021 is below; in case there is a similar situation, we use the following methods. == There are three chapters Chapter 1: Network modelling Chapter 2: MAC protocol Chapter 3: Routing + Lab: Simple experiments using laptops (individual) + simulations (group-wise) Marks Breakup -------------- Homework/Assignments/Group works Part A1: Group-wise assignment. 3 times (1 per chapter) -- 15 marks Part A2: Group-wise Q&A. 3 times (1 per chapter) -- 30 marks Part B1: Individual experiment + report Part B2: Group-wise simulations + report + demo Part B1 + Part B2 - 50 Marks Part C: Self-assessment + peer activities -- 5 marks Resit: Part A1 & A2 will not be repeated. Only Part B1 & B2 are allowed for the Resit this year, because of COVID19; and the projects should be done individually in Resit. == (More information will be given in the first class) Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
Permitted Materials during Tests	Different conditions for different test/exams. Conditions will be informed 1 week before the exams/test.

CESE4065	Advanced Practical I.o.T. and Seminar	5
Responsible Instructor	Dr. R.R. Venkatesha Prasad	
Contact Hours / Week x/x/x/x	2/0/0/0 and 0/0/0/2	
Education Period	1 4	
Start Education	1 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	fundamental understanding of wireless communications, familiarity with wireless communications and embedded systems and knowledge of Android programming/Python/C++/Matlab.	
Parts	Each seminar: 2x 45 minutes (2 parts) + 10 minute break	
Summary	his course is an involved hands-on course for self-motivated students. Students usually work in a group of two (max 3). Students are expected to have sufficient programming and hardware development skills. The course is project-oriented thus students are expected to deliver a working model with demonstrations. -- IMPORTANT NOTICE REGARDING ADJUSTMENTS DURING THIS COVID19 SITUATION The course will be offered ONLINE. Depending on the COVID19 situation there may be on-campus meetings/discussions [Safety of everyone is the highest priority]. -- This course is done in a group of two (max 3). The instructor/TA will help connect a potential groupmate in the first class using Brightspace/Feedback Fruits. Activities: How to in Groups? ----------------------------- Part 1: Seminar We guide the groups in selecting a paper. Students need to prepare slides and deliver a seminar using online tools; it is a group activity. For each seminar, all the students MUST be present. Absenting without permission will be taken seriously. Part 2: Project (1). Students can share the responsibilities if they can organize in such a way that one works on H/w and another on software so that they develop the project together. Meeting each other may be possible. However such face-to-face meetings should be done by strictly adhering to the COVID-19-related instructions from the university/government. (2). Students may exchange the components/boards within the group but carefully adhering to the 1.5m rule. The decision to do such cooperation is left to the judgement of the students and it is their responsibility. Please note safety is first. For Students not in NL or not able to be in Delft: (3). Instructor/TA would be offering an opportunity to do a hardware project if students can buy simple hardware like Arduino boards/Raspberry Pi, etc. They should try to use the method as in (1) above. (4). Instructor/TA may offer Algorithm/simulations/android programmes/Contiki-based networks or Open testbed projects. In these cases, online group activities are possible. NOTE: In some extreme cases, the Instructor/TA may allow one person to project after assessing the situation and after discussing it with the student. Supply of Hardware: ------------------- 1. Instructor/TA is able to provide take-home components (limited numbers) like Arduino boards and some sensors. We may provide some other instruments/sensors/boards depending on the selected project. These could be collected from TA while adhering to sanitization and social distancing rules. 2. Students may also use their own components.	
Course Contents	Course will be composed of a series of seminars related to the broad topic of the Internet of Things. Students will present their results on investigations regarding the possible extension of the ideas presented in the assigned papers.	
Study Goals	To be able to design components of Internet of Things and showcase an application or product through an implementation of a project. Specifically, to be able to bring entrepreneurial aspect of the project and also to be able to evaluate the project in depth. To be able to criticize and assess system-level components of the Internet of Things environment discussed in the scientific literature.	
Education Method	Seminar will be composed of (i) seminar presentation on a selected research paper (from top journals/conferences) presented individually by students and, (ii) work on a research project. Students will be provided with a list of projects that will be assigned to them. Project will be summarized in the form of a written report (report must include critical analysis). Within a project any hardware/software platform can be used and demonstrated. User experience/study, where applicable, also needs to be executed. Selected paper needs to be critically evaluated and a proposal to extend the assigned paper will need to be presented in a form of a presentation. Paper extension should focus on a system level idea. Presentation skills, thinking and reflection abilities are looked into carefully. The teams are composed of two (or three) students generally. NOTE: The total amount of work would be on the higher side of 150 hours; since this is an advanced course, students are expected to already have very good knowledge of hardware platforms, coding, and design environment. In case of lacking in some of these skills, we expect students to acquire them outside this budgeted 150Hrs. However, the number of hours of workload mentioned here is ONLY a guide, the efforts depend on the project, goals, and the collaboration with the team member(s).	
Literature and Study Materials	This is a project based course. Thus, Internet and other appropriate manuals (for chipsets, etc.) would be useful. For papers to read and present, we expect students to look into: Infocom, Mobicom, IPSN, Sensys, Sensapp, Mobihoc conference papers. Journals: IEEE Trans on Networking, Trans on Mobile computing, JSAC, etc. ACM Trans on CPS, etc.	

Assessment	Part 1: Assessment based on presentation quality, slides, Q/A, constructive criticisms, ideas for improvements, etc. Part 2: Project execution in a group, demonstration, Q/A and a report describing the outcome of the assigned project. Part 1: 30% of the whole marks; Part 2: 60% of the whole marks. In the assessment, a focus on the practicality and entrepreneurial aspect of the idea will be prevailing. A working model, demonstration, and a report, are expected. Individual Q/A after demonstrations will be part of the assessment. Part 3: Up to 10% marks would be awarded if the report is detailed above the minimum expected and the work is ready for a submission to a conference. Report will not have individual component, however, Q/A may have individual component. Please NOTE: There would be no resit since this is a hands-on course being executed in a team. As this is a project-based course, there is no official resit scheduled. Instead, an opportunity will be given to improve the work. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).
Tags	Circuits Group work Programming Project Software
maximum aantal deelnemers	Max 30 students, which translates to 10-15 groups (Groups consisting of 2 or 3 students).

CESE4075	**Supercomputing for Big Data**	**5**
Responsible Instructor	Dr.ir. Z. Al-Ars	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Students are expected to have a strong background in programming. Experience in Java, Scala or Python is a plus due to their importance in the big data ecosystem.	
Course Contents	Big data is one of the hottest technology fields today, used to describe large and complex data sets that are difficult to process using traditional data processing systems. In this course, we will introduce the students to the most important concepts of big data analytics and the available tools and systems used to manage and process it. In a series of labs, the students will be working with a number of different big data applications, addressing such aspects as implementing big data algorithms using Spark and Hadoop on high performance computational systems, as well as using higher-level languages and system management tools to guide and monitor the performance of these systems. The students will have access to a big data cluster to evaluate the effectiveness of their implemented solution in practice.	
Study Goals	By the end of this course, the students will be able to - use basic big data processing systems like Hadoop - implement parallel algorithms using the in-memory Spark framework, and streaming using Kafka - use libraries to simplify implementing more complex algorithms - identify the relevant characteristics of a given computational platform to solve big data problems - utilize knowledge of hardware and software tools to produce an efficient implementation of the application	
Education Method	Lab assignments and reading material provided via Brightspace and online on the CognitiveClass.ai site	
Assessment	The students will be evaluated based on the quality of their implementation and the report they submit describing their solution approach. The course has three lab assignments: L1, L2 and L3. The final grade of the course will be evaluated as Grade = 0.5 * L1 + 0.3 * L2 + 0.2 * L3 In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr.ir. A.J. van Genderen	

EE4740	Data Compression: Entropy and Sparsity Perspectives	5
Responsible Instructor	G. Joseph	
Instructor	Dr. N.J. Myers	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	[Theory] Good understanding of linear algebra and calculus (vector and matrix operations, gradient and Hessian), probability and statistics. [HW] Programming skills in Matlab/Python are required.	
Course Contents	The course covers a comprehensive overview of data compression and its connections to information theory and compressed sensing. The course content is presented in two parts: the first part covers data compression based on entropy, and the second part discusses compression based on sparsity. The first part introduces the fundamentals of the mathematical theory of information developed by Shannon. The course starts from probability theory and discusses how to mathematically model information sources. Entropy is established as a natural measure of efficient description length of optimally compressed data. Then, Shannons source coding theorem is discussed, followed by the entropy encodings of Huffman and arithmetic coding used in lossless data compression. The second part introduces the notion of sparsity and gives an overview of the area of compressed sensing. It starts with the classical techniques to solve underdetermined linear systems. Then, the core problem of compressed sensing, namely the l0 norm minimization with linear constraints, is introduced. The compressed sensing problem is motivated using example applications from wireless communication, control, and image processing. The l0 norm is shown to be NP-hard, and several classical approximate algorithms like l1 norm minimization, orthogonal matching pursuit, and thresholding algorithms are discussed. Building on the classical solutions, more advanced solution techniques are then covered. The first approach is the sparse Bayesian framework that introduces general statistical tools such as majorization-maximization and expectation-maximization. The next state-of-the-art approach is the deep learning framework based on autoencoders and generative adversarial networks. This part closes with the theory of compressed sensing. Here, the mathematical concepts of spark, null space property, and restricted isometric property are introduced. Using these concepts and properties, the theoretical performance guarantees of the standard compressed sensing algorithms are studied.	
Study Goals	At the end of the course, the student should be able to 1. Explain the basic notions and core theorems on the analysis and design of information representation. 2. Derive the limits to data compression and codes that meet the limits. 3. Understand the concept of sparsity and compressive sensing from an optimization perspective. 4. Debate the functioning of different sparse recovery approaches, namely convex optimization, greedy, thresholding, Bayesian, and deep learning techniques. 5. Discuss the theoretical underpinnings of sparse signal recovery and performance guarantees of compressed sensing algorithms. 6. Connect to recent literature on sparse signal recovery that builds upon the presented tools and techniques.	
Education Method	14 lectures; HWs; project presentations	
Course Relations	Although sufficiently independent, the course refers to EE4C03 Stat DSP, ET4386 Estimation and detection, EE4530 Applied convex opt.; SC 42140 Signal analysis & Learning-2D; SC 42150 Statistical SP	
Literature and Study Materials	1. T. M. Cover, J. A. Thomas, Elements of information theory, John Wiley & Sons, Inc, (2015) 2. S. Foucart, and H. Rauhut, A mathematical introduction to compressive sensing, Applied and Numerical Harmonic Analysis, Birkhäuser, New York, NY, (2013)	
Assessment	The assessment during the course comprises three quizzes (25%), a mini project (15%), and homework submission (10%). The course is closed by a project report, presentation, and Q&A (50%). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	

ET4169	Radar I: From Basic Principles to Applications	5
Responsible Instructor	Prof.dr. O. Yarovyi	
Responsible Instructor	O.A. Krasnov	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	ET4175, ET4173, ET4288, EE5020	
Expected prior knowledge	Basic knowledge of electromagnetism and electromagnetic waves (similar to those given in EE4C05 course), propagation and scattering of electromagnetic waves (some parts of EE4565 course) and basic knowledge of signal processing in detection and estimation.	
Course Contents	The course consists of three major parts. In Part 1 the fundamental issues of radar (signal and target models, range and velocity measurements with electromagnetic waves) are discussed. Part 2 of the course is focused on building an instrument (radar) that can sense objects remotely using microwaves. Attention is given to FMCW radar technology. In part 3 on examples of automotive radar and atmospheric remote sensing applications the radar signal processing and information extraction from radar signals are studied. Discussions on radar and remote sensing general principles are complemented by an analysis of modern radar systems designed and implemented at TU-Delft. The course includes a few practicum tasks, providing students with an experience in the experiments with real radars and in the processing of signals measured with a few types of radars. Disclaimer: the actual information related to this course content, education methods, and assessment may slightly change depending on the further development of the course and will be updated during the first lecture.	
Study Goals	The aim of this course is to introduce students to radar systems and active microwave remote sensing. The course will lay foundations for the further courses on radar and remote sensing.	
Education Method	The course consists of lectures, laboratory classes, home assignments and individual student's work.	
Literature and Study Materials	Lecture handouts; - Selected chapters of Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; - Selected chapters of D.M. Pozar, Microwave Engineering, 3rd ed., John Wiley & Sons, Inc., Hoboken, 2005; - Selected chapters of Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016; Additional sheets/ readers. Books: Mark A. Richards, James A. Scheer, William A. Holm, Principles of Modern Radar, SciTech Publishing, 2010; Fawwaz Ulaby, David Long et al., Microwave Radar and Radiometric Remote Sensing, 2016.	
Assessment	Written exam The final grade is calculated as: A student has to pass the written exam (exam grade higher than 5.5), then the final grade is calculated as: - 10% of averaged grades for homeworks (max 1 point); - 20% of averaged grades for practicum reports (max 2 points); - 70% of exam grade (max 7 points). In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Tags	Elektricity & Magnetism Signals Signals and Systems Technology Telecommunication	

IN4302TU	Building Serious Games	5
Responsible Instructor	Dr.ir. A.R. Bidarra	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Expected prior knowledge	For CS students: programming experience with some object-oriented language; experience with graphics, AI and/or some game engine(s) is a plus. For all students: though not compulsory, it may be convenient to have followed the course SEN9235 (Game Design Project), which is taught in the first quarter.	
Course Contents	Project-based interdisciplinary course, open to MSc students of all faculties (incl. LDE universities). In this project we put together a team of students with varying talents, backgrounds, and perspectives to achieve what none of them could do alone: to design and implement a serious game aimed at being applied in a real-world setting (educational, social, professional- and health-related, etc.). The emphasis is both on constructively fulfilling the game requirements, and on deploying the adequate technology for that purpose, solving a variety of possibly conflicting, technical challenges. Assignments for this course are provided by real-world commissioners (from startups to NGOs to research institutes), to whom the group will be reporting throughout the project. In previous years, various serious game prototypes originated in this course were actually developed and eventually deployed for their intended application; and many more were published and presented by students in international research venues (incl. IEEE CoG, FDG, ICIDS, GALA, ISAGA).	
Study Goals	At the end of the project, the student will demonstrate proficiency in the following aspects: o identifying and valuing the soft skills necessary to work effectively in an interdisciplinary team, integrating its members' talents in a variety of team roles o adapting with flexibility to the dynamic requirements of a complex external assignment o proactively translating feedback received into personal development steps o formulating research questions, identifying research contributions, and describing them in a scientific publication Additionally, the CS student will demonstrate proficiency in the following specific aspects: o identifying and deploying the most adequate game technologies for the given serious game domain and constraints o deepening programming skills while building a complex and large software system in an agile context	
Education Method	Project: teams work intensively as a small indie game studio, according to a tight planning. Also a few plenary sessions and/or lectures are held.	
Assessment	The final grade of the course consists of the following components: 1. Product grade (for the whole group) (weighting 50%). Based on both the game itself and the required documentation. 2. Process grade (individual) (weighting 45%). Including personal contribution, performance, attitude, and peer evaluation. 3. Final presentation (for the whole group) (weighting 5%). The course deliverables include writing a scientific paper and preparing it for actual submission to a conference on serious games and/or their application. Final grade calculation: 0.5 * Product grade + 0.45 * Process grade + 0.05 * Final presentation A passing final grade for the course can only be earned when for all components at least a 5.0 is earned, and the weighted final grade is at least a 5.8. Resit/ Repair opportunities: In case of an insufficient final result, repair options may exist in accordance with Article 17A Times and number of examinations, sub 1, of the Teaching and Examination Regulations 2024-2025, for: Final presentation: has no repair, as it is done in group. There is no repair possibility for the other components (product and process), since their assessment necessarily requires intensive teamwork throughout an extended development period. Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 2, sub 5).	
Co-Instructor	Dr. R. Marroquim	

RO47003	Robot Software Practicals	5
Responsible Instructor	J.F.P. Kooij	
Instructor	G.A. van der Hoorn	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basic programming experience (e.g. python or matlab) at the BSc level, familiarity with the use of functions, variables, and control structures (for-loops, if-statements, etc.).	
Course Contents	This course teaches students the basics for creating and collaborating on academic robotics software. It aims to give students sufficient knowledge and experience such that they can proceed learning more about robotics programming by themselves as needed. The course does this by providing an overview of key technologies, and letting students develop basic skills often used in real world robotic software development. The main objective is to obtain hands-on experience - beginners level - with Linux, Git & Gitlab, C++, ROS, and integrate with common external libraries such as OpenCV and PCL. This course does not teach new robotics theory (e.g. perception, planning, control, etc.) which is covered in other courses. The allotted 5 ECTS means that students are expected to invest approximately 11 hours/week outside of contact hours. Be aware that active participation in all the lab and peer-review assignments is required, hence it is not recommended to do this course if you cannot commit to a regular time investment over the whole education period.	
Course Contents Continuation	Linux: Although Windows is the most prevalent operating system on desktop PCs, Linux became very popular for embedded systems such as robots. Developing for embedded Linux systems is most easily done in Linux itself, and this course aims to familiarize students with the use of Linux on the desktop and terminal. It includes the following topics: OS architecture, File system, Shell, Scripting. Git/gitlab: All exercises will be submitted using Git, a widely used version control system to manage and collaborate on coding projects. During the course, students will practice basic tasks with git, such as solving code conflicts, and merging the work of lab partners. Using an issue tracker and giving constructive feedback to peers is also an important aspect that will be practiced. C++: C++ is one of the most relevant programming languages for robotics. Being an object oriented language brings great advantages compared to its predecessor C. This practical gives students practical knowledge on C++ programming. The course encompasses the following topics: Introduction to C++, common compilation tools, basic programming in C++, classes and objects. ROS: The Robot Operating System (ROS) is a flexible framework for writing robot software. It is a collection of tools, libraries, and conventions that aim to simplify the task of creating complex and robust robot behaviour across a wide variety of robotic platforms. During the course, students will learn to use existing ROS tools, create their own software components in C++, integrate them into an application, and test their solutions using a physics simulator. others: You will also get some practice using external libraries in your robotics project, such as OpenCV or PCL. OpenCV is a computer vision library designed for computational efficiency and with a strong focus on real-time applications. The Point Cloud Library (PCL) is an open-source project for point cloud processing. Note that this course is *not* about understanding the methods or theory on which these libraries are based, but rather on integrating the provided functionality in novel software components.	
Study Goals	--------------------------------------- Knowledge, insight, judgment and skills --------------------------------------- After completing this course, students will be able to 1. use the command line on a Linux system, perform basic file operations and job management tasks, and automate tasks with shell scripts; 2. use Git to commit and integrate incremental code changes together with a lab partner, and resolve any code integration conflicts; 3. use a Gitlab server to exchange code changes, and use its issue tracker to provide and receive feedback on each other's code; 4. solve programming tasks in C++, compile the code using Make, debug compile-time, link-time and run-time problems; 5. create new C++ ROS packages, use common ROS development tools, and utilize the ROS API in their C++ programs; 6. investigate and integrate external libraries into their own ROS components, such as OpenCV or PCL; 7. add code comments and basic documentation to their code, find and interpret documentation of external libraries; ------------------------------------- Transferable and interpersonal skills ------------------------------------- 8. develop evaluative judgments on the quality of your own work and of the performance of others (including peers) by implementing and providing feedback; 9. collaborate and coordinate with lab partner on coding tasks.	
Education Method	Lectures, which will introduce new course content (2 hours per week). Interactive Sessions will provide class wide feedback on past lab assignments, tutorials, quiz exercises, Q&A (2 hours per week) Practicum session, during which there will be TAs around to help you understand the lab assignments (2 hours per week) Robotics Software Practicals focuses on improving programming and collaborative software development skills. The main method to develop these skills is by gaining hands-on experience through lab assignments, and receiving feedback on the submitted work. Lectures will introduce the various topics, provide context and tips, and will discuss common pitfalls and solutions. Still, the nature of the assignments means that students are expected to invest sufficient time for self-study and practice outside the fixed lab and lecture contact hours. Lab assignment There are 4 obligatory lab assignments, each covering two weeks, namely 1. Linux, Git (Formative) 2. C++ programming (Formative) 3. ROS (Formative) 4. Final Assignment (Summative, demonstrates capacity to use Linux, Git, C++, and ROS). A manual will be provided for each lab assignment before the start of the first practicum session for the assignment. The manual will contain detailed instructions, guidelines, optional exercises, and an assignment that must be submitted to an online repository for your lab group.	
Literature and Study Materials	1. A Gentle Introduction to ROS, Jason M. OKane. http://www.cse.sc.edu/~jokane/agitr/ 2. Provided lab manuals, and lecture slides that will be made available on Brightspace	

Practical Guide	For the practicum assignments, bring a reasonably recent laptop with a x86-64 CPU (i.e. a normal" Intel or AMD CPU; but NOT the new MacBook Air with M2 chip, NOT a Chromebook, etc. These use different CPU architecture which means that the software we provide will not run! A MacBook with an Intel x86 CPU is fine). A laptop with an Nvidia GPU would make the robot simulations run even smoother. Please note the following lab rules: The first three lab assignments are made in groups of two, as you will learn to collaborate on code with your lab partner using gitlab. You can find a lab partner using the special Brightspace forum, but of course also by talking to others after the first lecture, or by looking for a Brightspace lab group with a single enrolled person. To enrol in the practicum, and therefore participate in the course, you and your partner *must* enrol on Brightspace in a lab group before the lab enrol deadline (this deadline will be announced in first lecture). If you plan to drop the course, inform the lab coordinator in time before the start of the next lab assignment, but while you are still enrolled, you should finish the ongoing lab assignment with your lab partner. The scheduled practicum sessions are intended for you to obtain help on the assignments from the TAs. Outside the practicum sessions, you can post questions on the TA forum on Brightspace. To submit your work, you and your partner must push your contributions to a provided gitlab repository. Note that using git and the gitlab webserver for code collaboration is part of the learning objectives of this course. You are responsible for testing the code that you hand-in via gitlab, so make sure that you have tested your work thoroughly, and that the correct version has been uploaded to your gitlab repository. For the first three lab assignments, there will be a peer-review task after the lab deadline has passed. You will learn during the course how to provide feedback, and how to use the gitlab issue tracker to manage issues and comments. The lectures will afterwards discuss positively peer-reviewed lab solutions with all students for additional feedback by a teacher. Active participation in each lab assignment is required to participate in the next assignment. Active participation entails that you submit your solution before the given deadline, and that you complete the required peer-review tasks. You may only use the gitlab code repository that the lab coordinator will create for your group. You are *NOT* allowed to put your coding homework on github.com, gitlab.com, bitbucket.com, google drive, dropbox, etc. (neither with public nor private settings!) TU Delft ethical guidelines apply to reports, code, etc. If your name is on it, you claim it is your work, do *not* copy parts of other student groups code. You are *not* allowed to share your code, solutions with others apart from your lab partner and TAs. Generally, do *not* show your groups work to other students. Plagiarism will be reported to exam committee.
Assessment	# Assessment overview Formative: First three lab assignments, peer-review tasks, discussion on common mistakes and solutions of past hand-in assignments during the lecture by teacher, multiple-choice practice questions. Summative: The final lab assignment, and digital exam. # Grade calculation You final course grade (minimum is a 6 to pass) determined by Final lab assignment (grade, weight 25%, minimum a 5.5 to pass), Digital exam (grade, weight 75%, minimum a 5.5 to pass) Any penalties (see Deadlines below) A resit will be available for the exam, but not the final lab assignment. # Lab assignments You must enroll for the obligatory lab assignments before the announced enrollment deadline (given during first lecture), so that the lab coordinator can fix the groups and setup everybodys code repositories in time. If you do not enroll, you cannot make the lab assignments, or make the exam. In the first three assignments you must collaborate with a lab partner. Your group will receive and give formative feedback through student peer-review (what goes well, what to improve). The fourth and final lab assignment is made individually, and you receive a grade (weights 25% for final grade), based on a rubric for the quality of the solution, code, documentation, and use of Git. This assignments requires you to apply all knowledge from the previous lab assignments to solve a well-defined robotics programming task. Your (groups) lab work is evaluated on the following three Lab Criteria (LC): - LC1: solution quality (does the solution solve the given task?) - LC2: code quality (are there any bugs or problems with the code?), and - LC3: documentation quality (is there a clear readme, does the code contain sensible comments?). # Deadlines, submission requirements, and penalties While most of the lab assignments are formative, your active participation is required and there are strict submission and peer-review requirements: - LC4: Each partner must fulfil each labs minimum submission requirements regarding use of Git, and Gitlab, file layout, and build instructions, before the lab deadline. Each assignments unique submission requirements will be listed in the corresponding lab manual. - LC5: You must participate in the peer-review tasks for the first three lab assignments, and complete these before the given peer-review deadlines. Failing to comply with these criteria will result in penalties on your final *course* grade: - After lab submission deadline: -1 grade point per day for not completing the assignments minimum submission requirements, with a maximum of -2 per assignment. Not enrolling in time before the lab automatically results in -2. - After peer review deadline: immediate -0.5 grade point for not fully completing the peer-review task (only applies to those enrolled for the lab). NB: - These criteria apply to both lab partners individually. If you comply with the criteria but your partner does not, only your partner will receive the penalty. - Penalties apply to your final course grade, so if you do not submit for one lab (-2 penalty) you could only complete the course with an 8 at best. - There are NO redos or extra assignment opportunities to remove a given penalty. You will have to redo the lab next year and comply with the deadlines then for a clean slate. # Exam There will be an individually-made digital exam, for which you will receive a grade (weights 75% for final grade, a minimum of 5.5 to pass). Questions are based on content discussed in the lectures, and on practical situations that you have encountered during the lab assignments. The exam will also require you to reflect on and refer to your own coding solutions for the final lab assignment, therefore to participate in the exam you *must* have participated in the final lab assignment. Practice questions will be provided before the exam so you can assess your own understanding first.
Enrolment / Application	Registration: To participate in this course, registration via MyTUDelft is not needed. More information regarding registration for courses, the procedure, and current deadlines can be found on

	https://www.tudelft.nl/en/student/me-student-portal/education/related/registration-courses-and-examinations/courses. To participate in midterms and endterms, (timely) registration for exams is obligatory. More information regarding registration for exams, the procedure, and current deadlines can be found on https://www.tudelft.nl/en/student/education/courses-and-examinations/examinations/registration-for-exams. *in some cases, there are entry requirements or additional criteria to be admissible to a course. These can be found in the OER/TER of the programme, https://www.tudelft.nl/en/student/me-student-portal/education/related/regulations.
Department	ME Department Cognitive Robotics

SEN9235	Game Design Project	5
Module Manager	Dr.ir. G. Bekebrede	
Instructor	Prof.dr.ir. A. Verbraeck	
Instructor	Dr.ir. G. Bekebrede	
Responsible for assignments	Dr.ir. G. Bekebrede	
Co-responsible for assignments	Prof.dr.ir. A. Verbraeck	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	The project is mandatory for the Advanced Modeling, Gaming and Design specialization, and open to students from all other faculties inside and outside the university.	
Expected prior knowledge	Bachelor degree	
Summary	The simulation game design project offers conceptual insights as well as hands on experience with simulation-gaming (SG). SG is an established field of practice with proven value for instance in the field of spatial and urban planning, ecology, engineering and design, public administration, business management, learning, research and consultancy. The staggering growth and success of the video gaming industry has triggered the interest in simulation games from paper based games to video games for learning and policy making - even more. Besides creativity and enthusiasm, there are no entry requirements. In this course, students acquire advanced knowledge in, and experience with, gaming-simulation for serious use, i.e. for policy- and decision support, organization and management and professional learning.	
Course Contents	Experienced speakers, from consultancy organizations (s.a. Accenture), game designers and universities, will give lectures on history, future, design, typology and facilitation of SG. (We will discuss the role of computers and video games, but emphasis is NOT on technology!) We will have ample opportunity to try out a number of (role-playing, board etc.) games that hold powerful messages about decision-making and management. Most of the work will be done in small teams, in which students will develop their own (non-digital) game to support a real-life or mock up case of policy/decision-making or organization and management. In a small group, you will also work on a scientific question related to game design.	
Study Goals	After completion of this course the student will have acquired knowledge and insights about: The history, backgrounds, key-concepts, formats and applications of simulation games. After completion of this course the student will be able to: 1. Design a (prototype) of a simulation game to be used for learning, research or intervention 2. Apply a game design cycle (or systematic design steps) 3. Define, conceptualize and construct the various game components 4. Facilitate simple simulation games	
Education Method	(Guest) lectures, workshops, game-play, game design, writing an essay Intensive workshops and group meetings Student group work	
Computer Use	Not likely; dependent on selected case. Possibility playing a computer game	
Course Relations	SEN9110 Simulation Masterclass (5 EC) Advanced Agent Based Modeling (5 ECTS) in4302 Building Serious Games (5 EC)	
Literature and Study Materials	To be announced (books and articles on gaming/simulation and game design). Reading materials will be made available via Brightspace.	
Assessment	Your final grade will be based on the following points: - Attendance and active participation during classes and workshops (required, no grade) - Grade for the game design assignment (group work 4-6 persons) counts for 80% of the final grade - Grade for the essay assignment (group work 2 persons) counts for 20% of the final grade - Facilitation of the warm-up game (same group as game design assignment) have to be done (required, no grade) Final grade of the course is 0,8 x grade game design + 0,2 x grade essay assignment, and student have to active present and facilitated a warm-up game. To sufficiently finish the course the weighted average have to be 5,75 or higher. Retake: - For practical reasons, there is no retake opportunity for the game design assignment - The essay can be partly repaired, the essay can be improved, maximum grade of the essay is a 6 - If a student can not join the assigned date for the facilitation of the warm-up game, we will look for an additional opportunity during the course weeks.	
Enrolment / Application	Registration through Brightspace enrollment	
Special Information	Geertje Bekebrede (g.bekebrede@tudelft.nl)	
Remarks	Attendance during the class meetings is required.	
Elective	Yes	
Tags	Challenging Design Group work Intensive Project	
Targetgroup	Master students of the modelling, simulation and gaming specialization(MSG) Elective course for students CoSEM, EPA, MOT, other faculties such as Architecture, Industrial Design, EWI, Civil Engineering etc.	
Category	MSc level	

WI4227-14	Discrete Optimisation	6
Responsible Instructor	Dr. F.M. de Oliveira Filho	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Specific prior knowledge: * Linear programming: simplex method, LP-duality * Basic knowledge of graphs/networks This is covered in the course TW2020 (Optimalisering) General knowledge: * Proficiency in reading and writing proofs * Sufficient mathematical maturity	
Course Contents	Discrete optimization is about the problem of finding a best solution among a (finite) set of feasible solutions. A well-known example is the traveling salesman problem, where we are asked to find a shortest tour in a graph visiting every node exactly once. This course is an introduction to the area, focusing on fundamental methods like linear programming and its theoretical framework (polyedra, integrality, etc.). The following topics will be covered: - Convex sets and their structure. - Polyhedra, cones, and their structure. - Linear programming, the simplex method. - Systems of linear inequalities, Fourier-Motzkin elimination. - Farkas's lemma, linear programming duality. - Several applications and examples: flows in networks, matrix games, bipartite matching, etc.	
Study Goals	By the end of this course, students will be able to - Recognize convex sets and their applications; - Apply the theory of convexity to concrete examples; - State the main results/theorems concerning polyhedra; - Model problems as linear programming problems; - Recognize total unimodularity and use it to prove integrality of polyhedra; - Construct the dual of a linear program.	
Education Method	Lectures	
Literature and Study Materials	The course is based on lecture notes and no specific book will be followed.	
Books	No specific book will be followed in this course. The following books contain most of the course's material: * A. Barvinok, A Course in Convexity, American Mathematical Society, Providence, 2002. * V. Chvatal, Linear Programming, W.H. Freeman and Company, New York, 1983. * W.J. Cook, W.H. Cunningham, W.R. Pulleyblank, and A. Schrijver, Combinatorial Optimization, John Wiley & Sons, New York, 1998. * A. Schrijver, Theory of Linear and Integer Programming, John Wiley & Sons, New York, 1986.	
Prerequisites	* Linear programming: simplex method, LP-duality * Basic knowledge of graphs/networks This is covered in the course TW2020 (Optimalisering) General knowledge: Proficiency in reading and writing proofs Sufficient mathematical maturity	
Assessment	The final grade of the course consists of the following components: - Written exam (100%) Resit/ Repair opportunities: In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Written exam: written resit Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4).	
Permitted Materials during Tests	No materials are permitted during the test (except pencil, ruler, eraser etc). In particular, calculators are not allowed.	

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